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Political segregation in the US workplace

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We estimate the extent of workplace political segregation in the United States by merging data covering over 45 million workers. We present four main findings. First, partisans are segregated by workplace. The average Democrat's coworkers are 11.7 percentage points (pp; 95% confidence interval (CI) [10.6, 12.8]) more Democratic than the average Republican's. After controlling for geography, industry and occupation, segregation is 2.9 pp [2.7, 3.1], comparable to analogously estimated gender segregation (2.8 pp [2.6, 3.0]). Second, segregation is largest among the politically active (political donors: 14.8 pp [13.2, 16.4] versus non-donors: 11.6 pp [10.5, 12.7]) and those with more market power (senior executives: 14.7 pp [13.4, 16.1]). Third, Republicans experience higher exposure to Democrats than vice versa: the average Republican's coworkers are 50% Democratic versus 32% Republican for the average Democrat. Fourth, political segregation has changed little over time (2012: 11.1 pp [10.0, 12.2] versus 2024: 11.0 pp [10.0, 12.0]).

Political partisans in the United States are increasingly segregated along geographic and social dimensions. Partisans sort into predominantly 'blue' or 'red' regions, states, cities and neighbourhoods[1–5]. Democrats are more likely to attend college[6,7], while Republicans are more likely to attend religious services[8]. Worshipers from both parties tend to share the pews with copartisans[9,10]. In marriage and courtship, partisans prefer copartisans and increasingly pass their political affiliation to their children[11–13]. Increased political segregation has coincided with rising 'affective polarization'—that is, the tendency of political partisans to dislike and distrust members of the opposing party[14,15]—raising concerns that declining interpartisan contact contributes to partisan animosity[16].

Despite growing evidence of political segregation across many domains of everyday life, political segregation at work remains understudied. This gap is substantial. For one, Americans spend far more waking hours at work than in any other activity[17]. Crucially, this time at work constitutes a rare opportunity for the sort of sustained, cooperative, cross-partisan contact that may reduce prejudice across societal groups[18–20]. In fact, the workplace may be one of the few domains that meets the criteria for prejudice-reducing intergroup contact inasmuch as it is frequently characterized by relatively equal status among participants, a common group goal that requires cooperation across members of different groups and the support of relevant authorities[21]. Almost by definition, coworkers cooperate in teams to achieve a common objective that has been set by a superior. Residential neighbours, by contrast, often interact only superficially and are less likely to interact at all if they are not copartisans[22]. Moreover, employment frequently constitutes an important source of social identity[23], meaning that workplaces with partisan diversity may constitute sources of cross-cutting cleavages which strengthen democratic norms[24].

Despite the apparent importance of workplace political segregation, research on this topic in the United States has focused on top managers or specific occupations or provided only small-scale experimental evidence that rank-and-file employees might sort by partisanship. Analyses of top managers reveal that boards of directors are increasingly politically homogeneous[25,26]. Experiments involving employee hiring suggest that rank-and-file employees prefer to work for companies that share their political ideology and/or partisanship[27–29] and that employers also prefer to hire politically like-minded candidates[30,31]. Given the absence of legal protection or strong norms against political discrimination in hiring, scholars have theorized that workplace segregation may be more pronounced than segregation based on legally protected categories such as gender[14]. Recent research suggests that, at least in Brazil, this is the case[31].

We address this gap by merging administrative data from a national voter file with a separate dataset of online employee profiles from Revelio Labs (LinkedIn). We merge these datasets via an ensemble approach that uses longstanding probabilistic approaches[32,33] as well as more recent large language model (LLM)-based techniques[34]. Full details of our source data and matching strategy can be found in the

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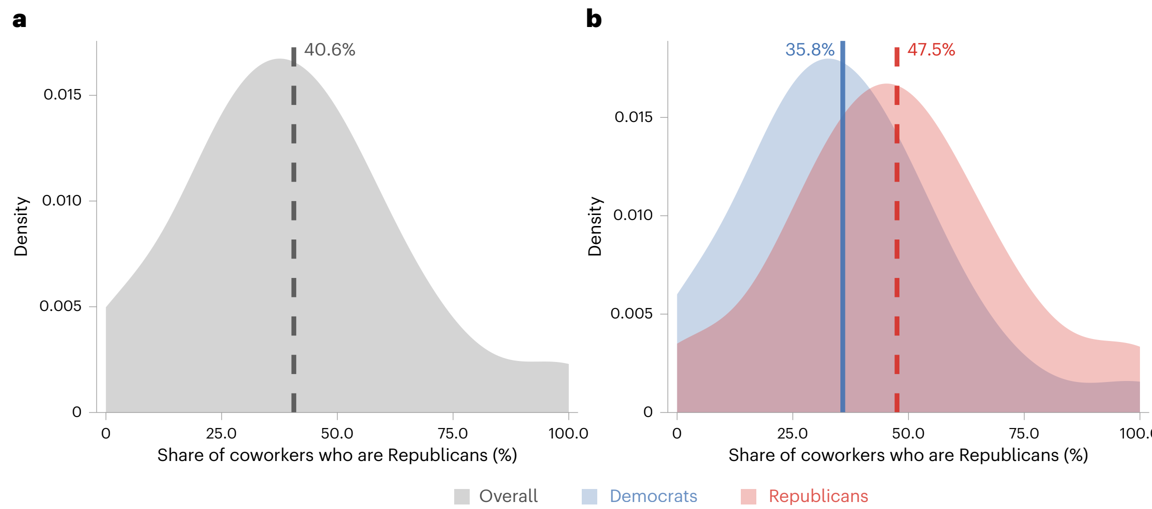

Fig. 1 | Nationwide distribution of coworkers that are Republican, overall and by focal worker partisanship. a, Overall distribution of the share of coworkers who are Republicans across all positions. **b**, The same distribution separately for Democratic and Republican focal workers, illustrating political segregation. Densities are estimated using a fixed bandwidth of 0.1; vertical lines indicate mean values (dashed grey in **a**; solid blue for Democrats and dashed red for Republicans in **b**). The sample is restricted to workers identified as Democrats or Republicans using imputed partisanship measures, and we include only workplaces with at least two employees. Republican coworker exposure is measured as the share of coworkers who are Republicans in the focal worker's establishment, calculated using two-party share measures. N = 37,200,614 positions for 30,965,666 unique workers. Supplementary Fig. 5 provides corresponding histograms.

Methods section below. Our full dataset contains over 45.3 million unique workers. The primary analysis is performed on a 2024 snapshot of 31.0 million unique workers, representing 30% of attempted matches, 19% of registered voters and 18% of the US workforce. To conduct temporal analysis, we also extend our 2024 matches backwards in time to construct a panel of position–years spanning 2012–2024.

Our coverage compares favourably to work in progress[35], which matches 34.5 million workers using a different matching approach and source data (versus 45.3 million in our approach), and which conducts most analyses on a 20.3 million worker–coworker sample (versus 31.0 million in ours). Our coverage also surpasses the 7.8% coverage of the Brazilian labour market achieved by ref. 31. When benchmarked against a nationally representative sample, our data compare well in terms of partisanship and gender. As is inherent to online profile data, our sample is moderately skewed towards white-collar industries and occupations. Despite this, our data constitute a meaningful improvement on past attempts to characterize worker partisanship that rely on political donation records[36–38]. Not only do very few workers donate[39,40], but donors also tend to be exceptionally old, white and educated[41,42] and thus are probably far more unrepresentative of the overall employee population.

We present four key findings regarding political segregation in the US labour market. First, partisans are segregated by workplace. The average Democrat works in an environment that is 11.7 percentage points (pp) more Democratic (P < 0.001, 95% confidence interval (CI) [10.6, 12.8]) than the workplace of the average Republican. We also compare the extent of political segregation that individuals experience at work with the residential political segregation they experience at home and find significant spatial variation. In some metropolitan areas, workers are more likely to encounter members of the opposing party at work, while in others, they are more likely to encounter outpartisans at home. After netting out sorting by occupation, industry and geography, this political segregation is comparable and by some measures larger than estimates of gender segregation. Using fixed-effect regressions, we show that substantial political segregation persists even within narrowly defined labour markets. Our most restrictive specification finds that even controlling for gender, race/ethnicity, home census tract, industry and occupation, a Republican is still likely to have ~3 percentage points more Republican coworkers relative to a Democrat.

Second, political segregation varies by worker characteristics. Notably, segregation is greater among workers who participate more in politics (for example, by voting in Presidential primaries and/or making political donations). Segregation is also higher among workers who plausibly enjoy greater market power, including white workers, younger workers, workers in occupations that require greater training and preparation, and those in senior management roles. These patterns suggest that workers who care more about politics and have more employment options are more likely to sort themselves into politically homogeneous workplaces.

Third, we find that partisan-based workplace segregation has increased only very modestly between 2012 and 2024. New hires show some tendency to increasingly join firms where a higher share of coworkers share their partisanship.

Fourth, Republicans experience significantly higher exposure to Democrats at work than vice versa. This asymmetry in outpartisan exposure is driven in part by the fact that Democrats comprise a greater share of workers than Republicans in the population, due in part to age differences between the parties (younger voters in peak employment years are more likely to be Democrats; Republicans hold an advantage among older voters who are less likely to be employed)[43]. We demonstrate that this asymmetry is not simply an artefact of our sampling methodology (see Extended Data Table 1). While the construction of our sample does overrepresent white-collar industries and better-educated workers, we show that the partisan composition of our sample closely matches the population of registered voters who are employed in the United States. This baseline difference in employment between Democrats and Republicans is further magnified by political segregation.

We close by discussing how the data and measures we present here can inform future work into the causes and consequences of workplace political segregation.

Results

Main estimates

This paper examines the workplace experiences of individual employees rather than the political composition of companies as a whole. Specifically, we focus on the extent to which the coworkers of a focal

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employee share this focal employee's partisanship. Given this focus, our methodology differs from existing work on company-level characteristics, such as political diversity within boards[26] or manager–employee dyads[31]. Instead, our approach most closely relates to work in ref. 1, which captures geographic residential segregation using the partisanship of residential neighbours. We adopt a similar approach: for each worker, we conceptualize coworkers as the set of all positions that share a workplace—that is, are at the same company and located within the same Metropolitan Statistical Area (MSA) (excluding the focal worker).

To define a worker's politics, we begin with voters who are formally registered with either the Democratic or Republican party, and extend our analysis by imputing the likely partisan 'lean' for registered independents, third-party members and registered voters who decline to affiliate with any political party. Our approach to imputing partisan lean is motivated by research which suggests that true independents are rare; nearly all nominally 'independent' voters consistently favour one of the two major parties[44–46]. Our approach to imputation follows ref. 1. We first assign partisan lean to unaffiliated voters with a recent past history of voting in either the Democratic or Republican primary. For voters registered with a third-party with a clear left–right ideology, we assign them as 'leaners' to the appropriately ideologically aligned party. (For example, the Green Party is considered to lean Democratic while the Libertarian Party is considered to lean Republican. We follow the party classifications of ref. 1.) For the remaining voters, we use a Bayesian modelling approach that incorporates both geographic and demographic imputation to impute voters' likely partisan lean.

In our main analysis, we focus on the most up-to-date snapshot of positions as of the time of our study. This sample comprises all positions that were active at any point from the start of 2024 through April 2025. Additional details on the source data, merge strategy and imputation methodology are provided in the Methods.

Formally, we define the share of Republican coworkers for a given position as:

$$\text{ShareRep}_{i,e,m} = \frac{\sum_{j=1, j\neq i}^{n_{e,m}} \mathbf{1}(p_j = \text{Rep})}{n_{e,m} - 1} \quad (1)$$

where $\text{ShareRep}_{i,e,m}$ is the share of coworkers for a worker's focal position i at employer e and MSA m that are Republicans (including Republican leaners). (All analyses that refer to Democrats or Republicans include leaners unless otherwise specified.) The term $n_{e,m}$ is the number of positions in firm e in MSA m, and $p_j \in \{\text{Dem}, \text{Rep}\}$ is the partisanship of coworker j, such that $\mathbf{1}(p_j = \text{Rep})$ is equal to one if coworker j is a Republican and zero if they are a Democrat.

Because a focal worker must have at least one coworker for our measure to be defined, our main analysis focuses on workplaces with at least two employees; our results are robust to a range of higher number-of-coworker inclusion thresholds. Figure 1a presents the kernel density plot of the overall distribution of the share of coworkers that are Republican, illustrating that the average worker (regardless of their party) experiences fewer Republican (~40.6%) and more Democratic (~59.4%) coworkers.

Figure 1b illustrates the degree of political segregation, showing that the average share of Republican coworkers varies significantly according to the partisanship of the focal worker. The mean Democrat's coworkers are 35.8% Republicans, while the mean Republican's coworkers are 47.5% Republicans—an 11.7 percentage point difference ($P < 0.001$, 95% CI [10.6, 12.8]). All reported P values are from two-tailed tests.

Geographic variation in workplace segregation

Having established that political workplace segregation exists (Fig. 1), we next examine how this segregation varies geographically. Our approach involves two key steps that allow us to separate workplace-specific sorting from residential sorting patterns.

First, for each of the 366 metropolitan areas in our sample, we calculate each level of raw workplace segregation using the following specification:

$$\text{ShareRep}_{i,e,m} = \alpha_m + \beta_m \text{Rep}_{i,e,m} + \epsilon_{i,e,m}, \quad (2)$$

separately for each metropolitan area m. The coefficient β_m measures the raw partisan gap for each worker's coworkers in each MSA; α_m is an MSA-level intercept; $\text{Rep}_{i,e,m}$ indicates whether the worker is Repulican; $\epsilon_{i,e,m}$ is an error term. Figure 2a maps these coefficients, revealing that while workplace segregation is a ubiquitous feature of the American workplace, its intensity varies significantly by region. The highest levels are heavily concentrated in the Deep South, with MSAs such as Montgomery, AL, and New Orleans, LA, exhibiting the most pronounced sorting. High levels of segregation are also notable in the Industrial Midwest. In contrast, the lowest levels of sorting are found in the Great Plains and Mountain West.

The raw workplace segregation shown in Fig. 2a, however, probably reflects two factors: (1) workplace-specific sorting and (2) the fact that Republicans and Democrats live and work in different neighbourhoods within an MSA. To isolate workplace-specific patterns, we ask 'How much workplace segregation would we expect in each metro area given its level of residential segregation?' To answer this, we first formally measure residential segregation for each metro area (m) using the following model:

$$\text{CensusTractRepShare}_{i,m} = \alpha_m^{\text{res}} + \delta_m \text{Rep}_{i,m} + \epsilon_{i,m}^{\text{res}} \quad (3)$$

where $\text{CensusTractRepShare}_{i,m}$ is the Republican share of voters in individual i's residential census tract (estimated using the full voter file) and $\text{Rep}_{i,m}$ is an indicator variable for whether individual i is a Republican. The resulting coefficient, δ_m, captures the level of residential segregation for that MSA.

We then predict workplace segregation from residential segregation using:

$$\beta_m = \gamma_0 + \gamma_1 \times \delta_m + \nu_m, \quad (4)$$

where β_m is the raw workplace segregation from equation (2), δ_m is the residential segregation coefficient from equation (3), γ_0 is an intercept and γ_1 is an estimated coefficient for δ_m. The residuals from this model, ν_m, represent the 'net' workplace segregation that is unexplained by residential patterns. Negative residuals indicate that workers experience less political segregation at work than in their residential areas. Figure 2b maps these residuals. The data suggest that net workplace segregation is systematically related to the economic and institutional character of an MSA. Large metropolitan areas consistently show negative residuals, with all of the largest 20 MSAs exhibiting less workplace segregation than their residential patterns would predict. Major 'knowledge economy' and technology hubs such as San Jose (CA), San Francisco (CA), Seattle (WA), Austin (TX) and Boston (MA) exhibit negative residuals, suggesting that these areas have more political integration at work than within residential neighbourhoods. Taken together, these analyses demonstrate that while raw workplace segregation follows broad regional contours, the underlying drivers are probably also tied to the specific economic and institutional characteristics of the metropolitan area.

Political segregation across industries and occupations

The geographic variation in workplace segregation documented in Fig. 2 suggests that broader political sorting processes influence where partisans work. However, workplace segregation may also reflect the fact that Republicans and Democrats systematically sort into different types of work—that is, different industries and occupations—even if they live in the same areas.

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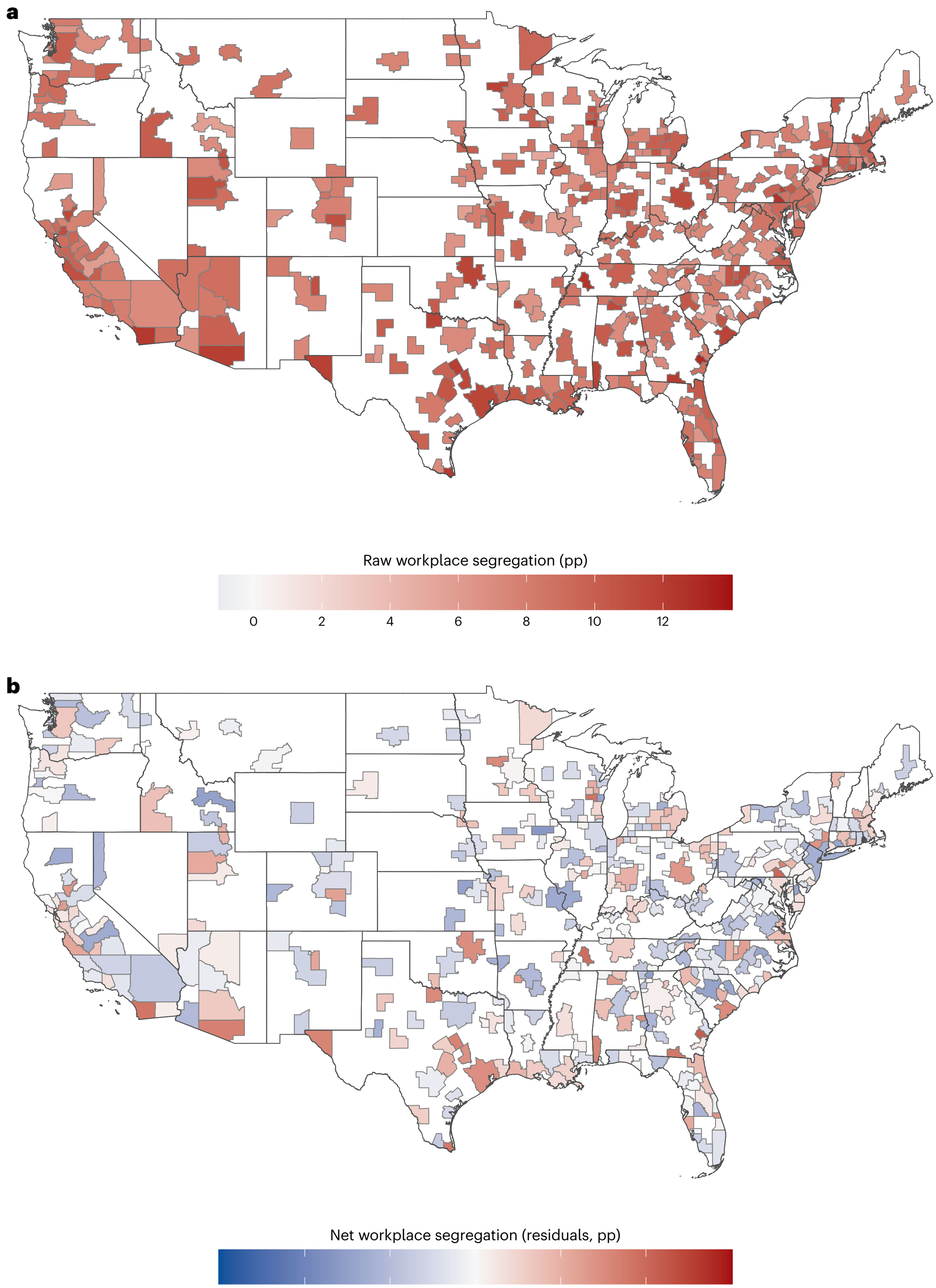

Fig. 2 | Workplace political segregation across metropolitan areas. a, MSA-specific coefficients β_m from equation (2)—the raw difference in Republican coworker exposure between Republican and Democratic workers in each metropolitan area. **b**, Residuals v_m from equation (4)—workplace segregation that cannot be explained by existing census-tract measures of residential segregation (equation (3)). The diverging colour scale runs from blue (negative values; less workplace segregation than predicted by residential patterns) through white (zero) to red (positive values; more workplace segregation than predicted). N = 37,200,614 positions for 30,965,666 unique workers across 366 metropolitan areas; the contiguous US map displays 363 of these MSAs. See Supplementary Tables 1 and 2 for complete MSA-level coefficients and residuals.

To examine how partisanship varies across different types of work, Fig. 3 presents ridge plots showing the distribution of Republican coworker exposure across selected industries and occupations. Figure 3a shows the overall distribution of Republican coworker exposure, which is equivalent to Fig. 1a. Figure 3b shows selected industries spanning the political spectrum, revealing dramatic differences in partisan exposure by industry and occupation. For example, workers in museums and environmental organizations experience, on average, only

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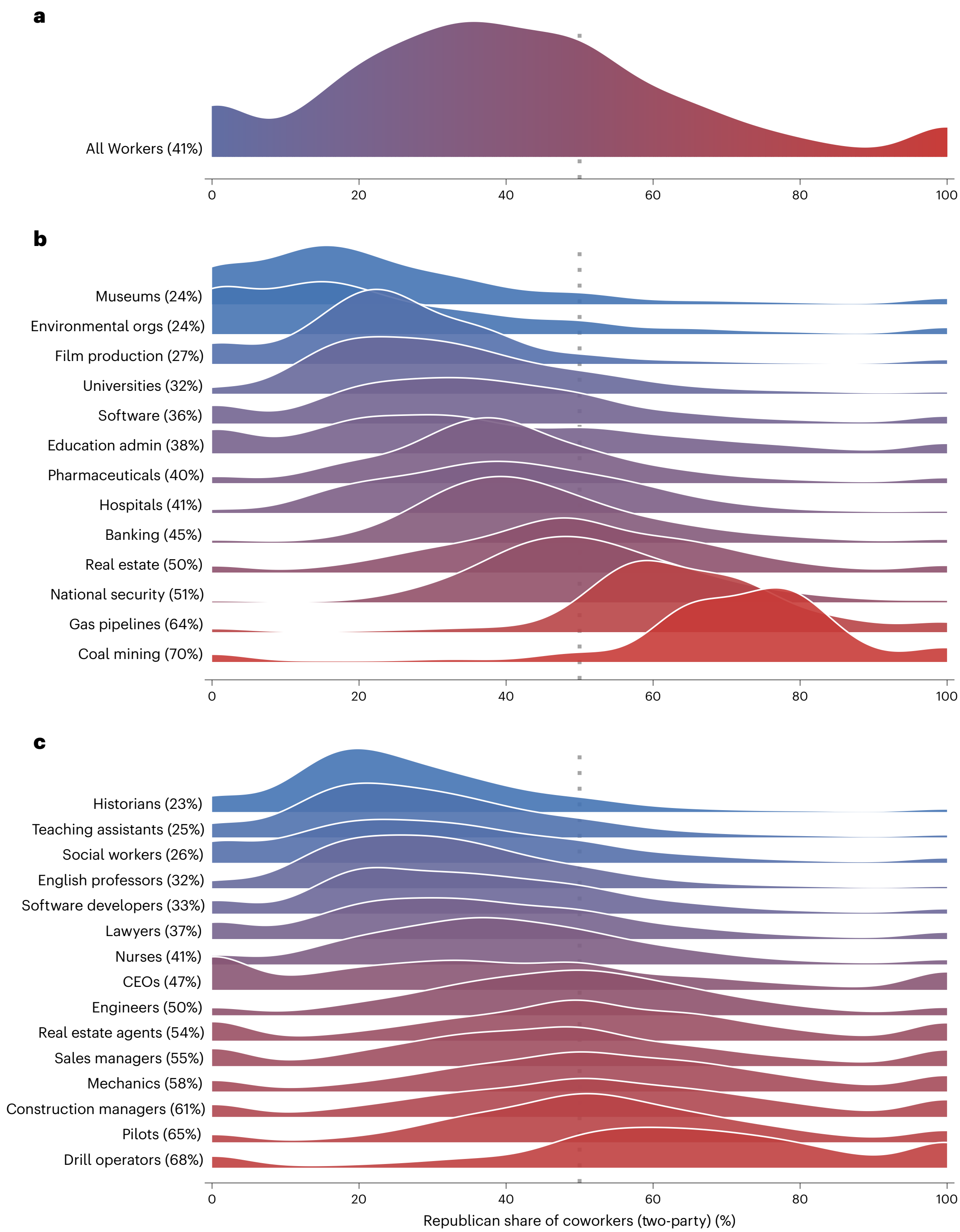

Fig. 3 | Ridge plots showing partisan compositions across occupations and industries. a–c, Ridge plots show the distribution of Republican coworker share (two-party) across selected industries (**b**) and occupations (**c**), with **a** showing the overall distribution across all observations. The fill colour encodes the same Republican coworker share shown on the *x* axis: blue indicates a low Republican share (Democratic-leaning workplaces) and red indicates a high Republican share (Republican-leaning workplaces), with intermediate values shown as graded purples. Vertical dotted line indicates political balance (50% Republican). Kernel density estimates use a fixed bandwidth of 0.05. N = 37,200,614 positions for 30,965,666 unique workers. Orgs, organizations; admin; administration.

23% Republican coworkers. In contrast, those in gas pipelines and coal mining experience 64% and 70% Republican coworkers, respectively. Figure 3c shows selected occupations with similarly pronounced political sorting. Historians and social workers experience, on average, only 22% and 26% Republican coworkers. Meanwhile, pilots and drill operators experience 65% and 68% Republican coworkers respectively. Ridge plots showing the distribution of Republican coworker exposure across all 2-digit North American Industry Classification (NAICS) industry codes and 2-digit Standard Occupational Classification (SOC) occupation codes are presented in Supplementary Figs. 7 and 8, respectively.

This occupational and industrial sorting represents a key mechanism through which partisans may experience workplace segregation even within the same geographic area. Workers choosing careers aligned with their political values or being drawn to industries and occupations with compatible worldviews would generate partisan workplace clustering independent of residential sorting patterns.

Accounting for geography, occupation and industry

Having documented political segregation across geography (Fig. 2), industries and occupations (Fig. 3), we now turn to our central analytical question: do workers sort into politically homogeneous workplaces beyond what would be expected from their residential, occupational and industrial choices? Our empirical strategy involves estimating increasingly restrictive specifications of equation (5), progressively adding fixed effects to examine political sorting net of other types of sorting. We estimate:

$$\mathrm{ShareRep}_{i,e,m} = \alpha + \beta \mathrm{Rep}_{i,e,m} + \boldsymbol{\lambda_j} + \epsilon_{i,e,m} \quad (5)$$

which is nearly identical to equation (2) but adds a vector of fixed effects, $\boldsymbol{\lambda}_j$.

We additionally re-estimate equation (5) in two ways. First, to account for the overrepresentation of white-collar workers in our sample, we fit a version of equation (5) that weights observations by occupation. Our weights use Bureau of Labour Statistics Occupational Employment and Wage Statistics (BLS-OEWS) to adjust our sample to match the national distribution of workers across occupations, thus accounting for potential sample selection bias arising from differential rates of voter registration, LinkedIn profile availability and matching success. We discuss both our sample benchmarking exercises and the weighting strategy in more detail in Methods.

Second, we also benchmark our estimates of political sorting against analogously estimated measures of gender segregation (gender is obtained from voter registration records where available, or imputed by L2 using modelling techniques based on first names). This analysis is run on our sample of 37.2 million positions held by 31 million workers. We repeat the analysis but replace (1) the outcome variable 'ShareRep' with the variable 'ShareWomen', which, for the focal position, is the share of women coworkers and (2) the explanatory variable 'Rep' with 'Woman', which is a dummy equal to one if the focal worker is a woman.

In line with Fig. 1, the raw estimate for political segregation in Fig. 4 shows that political segregation is 11.7 percentage points ($P < 0.001$, 95% CI [0.106, 0.128]) unweighted and 10.7 percentage points ($P < 0.001$, 95% CI [0.087, 0.128]) when weighting to BLS. This is somewhat lower than the raw gender segregation estimate, at 15.6 percentage points ($P < 0.001$, 95% CI [0.152, 0.161]). However, the patterns diverge markedly as we account for industry and occupation. Occupation fixed effects reduce political segregation modestly to 9.4 percentage points BLS weighted ($P < 0.001$, 95% CI [0.073, 0.115]) from 10.2 percentage points unweighted ($P < 0.001$, 95% CI [0.091, 0.112]) while dramatically reducing gender segregation to 8.4 percentage points ($P < 0.001$, 95% CI [0.082, 0.085]), indicating that occupational sorting explains much more of gender than political workplace segregation. Industry fixed effects show a similar but less pronounced pattern (political: 8.6 pp BLS weighted, $P < 0.001$, 95% CI [0.066, 0.106]; 9.6 pp unweighted, $P < 0.001$, 95% CI [0.086, 0.106]; gender: 7.2 pp, $P < 0.001$, 95% CI [0.071, 0.073]). We also include geography fixed effects based on workers' residential address. Census-tract fixed effects reduce political segregation substantially to 5.3 percentage points BLS weighted ($P < 0.001$, 95% CI [0.047, 0.059]), 5.9 percentage points unweighted ($P < 0.001$, 95% CI [0.055, 0.062]) while barely affecting gender segregation (15.5 pp, $P < 0.001$, 95% CI [0.151, 0.159]), demonstrating that fine-grained geographic sorting plays a much more important role in political versus gender workplace segregation. The use of census-tract fixed effects allows us to compare workers who live in the same neighbourhood but work for different employers, thereby isolating workplace-specific political sorting from residential segregation patterns.

In Fig. 4b, we combine occupation, industry and geography fixed effects additively. Both types of segregation decline substantially, with BLS-weighted political segregation (2.4 pp, $P < 0.001$, 95% CI [0.021, 0.027]) now lower than gender segregation (5.3 pp, $P < 0.001$, 95% CI [0.052, 0.054]). Using interactive occupation-industry-census-tract fixed effects yields similar results: political segregation (2.9 pp, $P < 0.001$, 95% CI [0.027, 0.031]) is about the same as gender segregation (2.8 pp, $P < 0.001$, 95% CI [0.026, 0.030]).

Our strictest specifications (Fig. 4c) control for demographics. Models examining political segregation include gender and race/ethnicity fixed effects. Models examining gender segregation include race/ethnicity and partisanship fixed effects. The additive specification shows BLS-weighted political segregation (2.1 pp, $P < 0.001$, 95% CI [0.019, 0.023]) below gender segregation (5.3 pp, $P < 0.001$, 95% CI [0.051, 0.054]), while the interactive specification shows similar levels: 3.1 percentage points ($P < 0.001$, 95% CI [0.025, 0.037]) versus 2.7 percentage points ($P < 0.001$, 95% CI [0.025, 0.029]). This suggests that after accounting for occupational, industrial, geographic and demographic sorting, political and gender workplace segregation are of similar magnitude. Importantly, the BLS-weighted results shown in Fig. 4 are substantively similar to unweighted estimates, assuaging sample selection concerns.

These fixed effects regressions quantify political sorting and provide descriptive insights into how political workplace segregation relates to other types of labour market sorting. However, it is important to note that these analyses do not isolate taste-based sorting along political lines. The analyses shown here focus on segregation within existing labour markets defined by geography, industry and occupation, but it is also probably the case that labour market entry is endogenous in this context. Specifically, to the extent that workers' choices about where to live and what occupations and industries to pursue are themselves at least in part motivated by politics, our observed estimates in these fixed-effect analyses may be conservative underestimates of the true extent of taste-based sorting motivated by a preference for copartisan coworkers. In the discussion, we further discuss how these descriptive patterns might inspire future work that might further isolate the causes and consequences of political sorting across workplaces.

Specification curves and sensitivity analysis

We made several methodological choices in calculating our estimates. We made these choices in light of our goal of understanding how workers experience their coworkers' partisanship in their workplaces, and we discuss these choices in more detail in Methods. To demonstrate that our results are not contingent on these choices, we conduct a multiverse analysis (specification curve) that systematically varies analytical specifications across multiple dimensions[47]. In the main analysis, we examine the two-party share while imputing Democratic or Republican partisan affiliation for workers who identify as independent, those affiliated with third parties, or those who do not disclose their partisanship, following ref. 1. In Supplementary Fig. 9, we show that the results are qualitatively similar using measures of partisanship that rely on non-imputed voter registrations. In addition, we demonstrate that our findings are robust to excluding workplaces with fewer than 2, 10 or 50 matched workers. We also show that our results are robust to clustering standard errors at the MSA and employer level or simply estimating robust (non-clustered) standard errors. This multiverse analysis encompasses both our baseline specifications (equivalent to equation (2)) and full (additive) fixed-effects models (equivalent to equation (5)), with results presented in Supplementary Fig. 9.

We also conduct sensitivity analysis around our decision to measure coworkers in terms of those that share a focal worker's employer–MSA. While some companies (for example, chain retailers) may have many locations within the same MSA, the population of workers within a given MSA represents an estimate of the population of workers with

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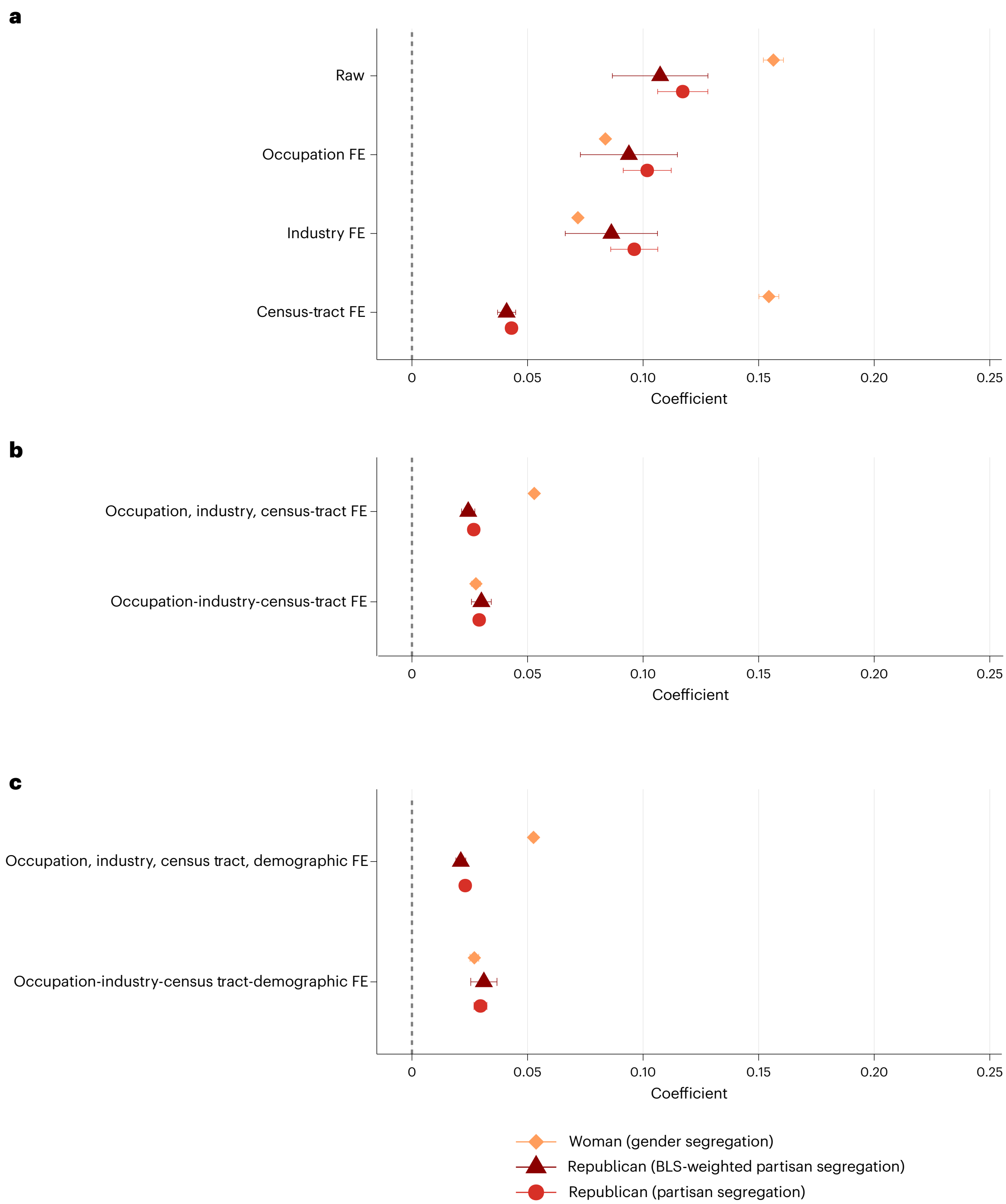

Fig. 4 | Estimated political segregation benchmarked against gender segregation. Point estimates are ordinary least squares (OLS) coefficients from versions of equations (2) and (5) that differ in terms of whether they characterize political or gender workplace segregation and the fixed effects they include. All BLS-weighted estimates are weighted using BLS-OEWS data to adjust for sample selection bias. Bars represent 95% confidence intervals; standard errors are clustered by MSA. **a**, Individual fixed effects (FE). **b**,**c**, Census-tract fixed effects. In the saturated models (**c**), demographic fixed effects are included as follows: political segregation models include gender fixed effects, while gender segregation models include partisanship fixed effects. N = 37,200,614 positions for 30,965,666 unique workers. All estimates use the fixest package in R. See Supplementary Table 3 for full regression results.

whom a focal worker could plausibly interact. Workers at the same company but in different MSAs are less likely to interact. To address potential measurement error that may arise when a firm has multiple establishments in a given MSA, we conduct sensitivity analysis restricting our sample to firms less likely to have multiple locations in a given MSA (employers with low overall number of workers and professional services firms); our results are robust to these restrictions (see Supplementary Fig. 10).

Heterogeneity

Next, we examine heterogeneity in segregation by worker characteristics (see Fig. 5). First, we examine heterogeneity in terms of political engagement, which reveals some of the strongest and most consistent patterns in our analysis. Political activity shows a clear gradient in workplace segregation: non-voters in the 2020 general election exhibit lower levels of segregation (10.4 pp, $P < 0.001$, 95% CI [0.092, 0.116]) compared with general election voters (11.9 pp, $P < 0.001$, 95%

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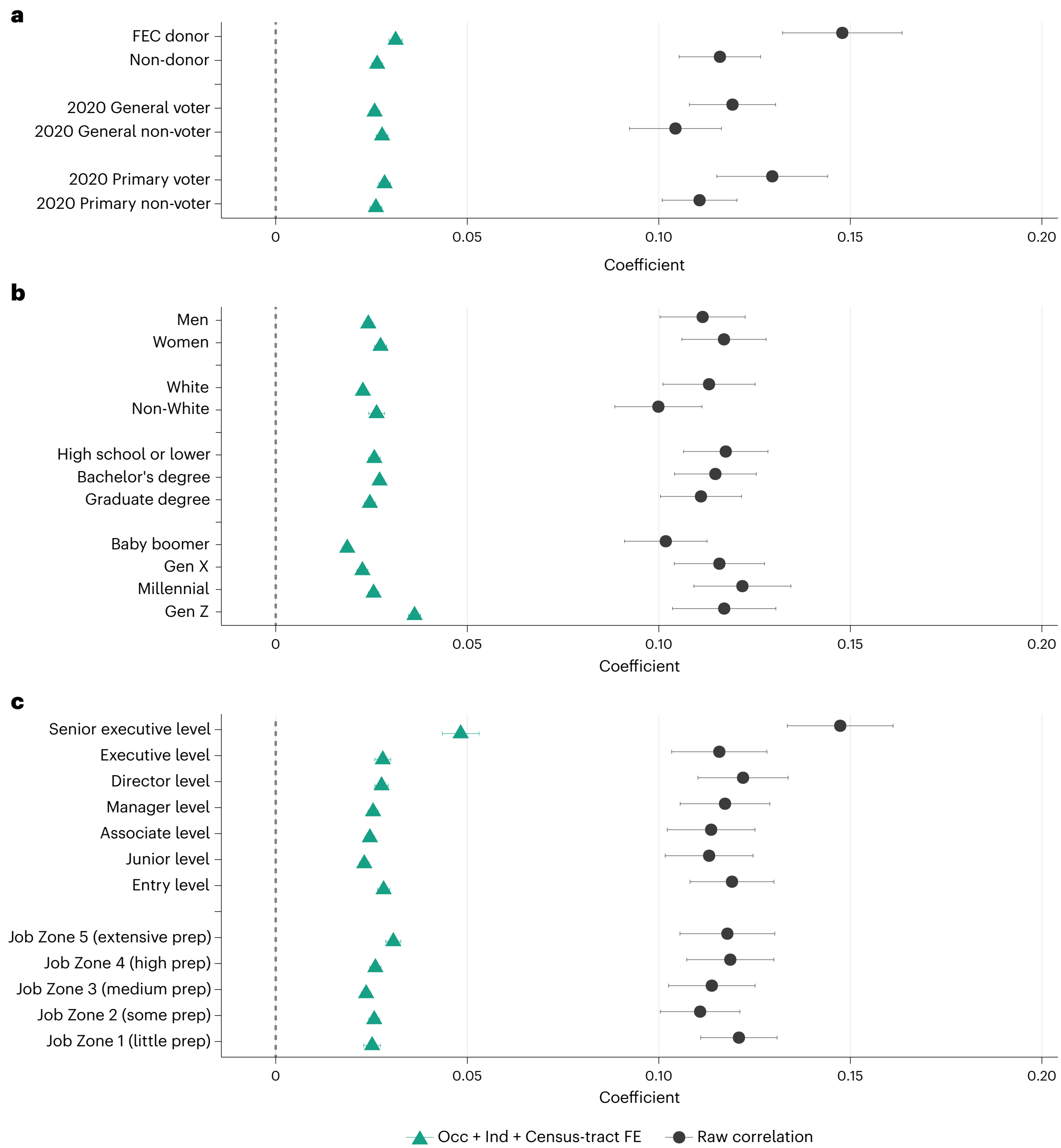

Fig. 5 | Examining heterogeneity. a–c, Comparison of sorting across levels of political engagement (**a**), by demographics (**b**) and by job seniority and experience (**c**). Each point represents an OLS coefficient estimating the association between being a Republican and the share of Republican coworkers in the employer–MSA, estimated separately for each subgroup. Colour and shape distinguish the two model specifications: dark grey circles show raw correlations (no fixed effects), and green triangles show estimates from models with occupation, industry and census-tract fixed effects. Error bars show 95% confidence intervals. N = 37,200,614 positions for 30,965,666 unique workers. All coefficients and standard errors are reported in Supplementary Table 4.

CI [0.108, 0.131]), while primary voters—who represent the most politically engaged segment of the electorate—display substantially higher segregation (13.0 pp, P < 0.001, 95% CI [0.115, 0.144]). Most strikingly, those who have made donations above the Federal Election Commission (FEC)'s US$200 reporting threshold show the highest segregation levels of any group we examine (14.8 pp, P < 0.001, 95% CI [0.132, 0.164]), with raw segregation coefficients 27% higher than for non-donors (11.6 pp, P < 0.001, 95% CI [0.105, 0.127]). This clear gradient from non-voters to donors suggests that political engagement itself, rather than simply partisan identity, drives workplace sorting behaviour. These patterns are consistent with the idea that those who are most politically active place the greatest premium on working alongside those who share their partisanship and may be more willing to factor political considerations into their employment decisions.

Second, we examine heterogeneity by worker demographics and experience. Among demographic characteristics, women experience slightly higher segregation than men (11.7 pp, P < 0.001, 95% CI [0.106, 0.128] versus 11.1 pp, P < 0.001, 95% CI [0.100, 0.123] in raw estimates), while white workers exhibit somewhat higher segregation than non-white workers (11.3 pp, P < 0.001, 95% CI [0.101, 0.125] versus 10.0 pp, P < 0.001, 95% CI [0.089, 0.111])—a pattern that may reflect differences in labour market flexibility and employment options. We observe a clear education gradient: workers with high school education or lower show the highest raw segregation levels (11.8 pp, P < 0.001, 95% CI [0.107, 0.129]), while those with bachelor's (11.5 pp, P < 0.001, 95% CI [0.104, 0.125]) and graduate (11.1 pp, P < 0.001, 95% CI [0.100, 0.122]) degrees show somewhat lower segregation. There is a similarly clear gradient across generations. Baby boomers exhibit the lowest

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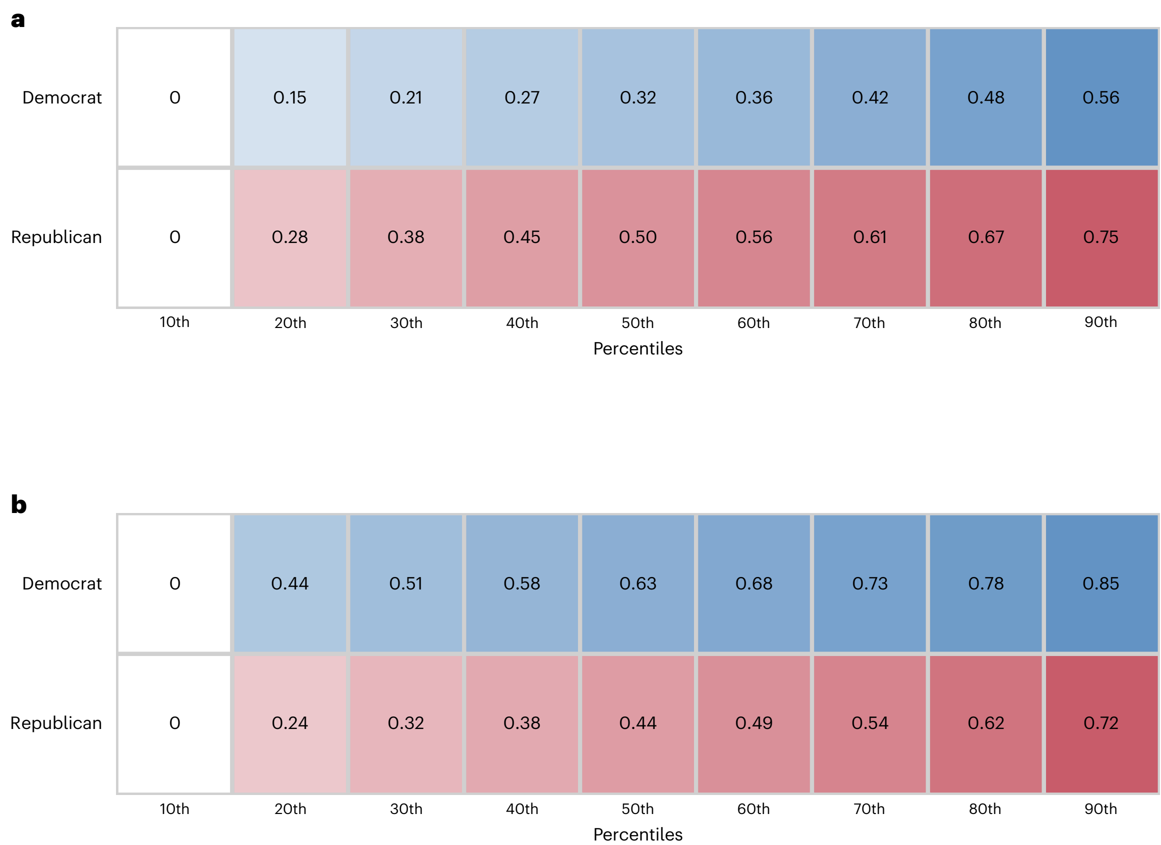

Fig. 6 | Distribution of in-party and out-party exposure, by political affiliation. **a**,**b**, Heat maps display the share of out-party (**a**) and in-party (**b**) exposure by decile. Values shown are mean exposure rates within each decile. Out-party exposure measures Republican coworker share for Democrats and Democratic coworker share for Republicans. In-party exposure measures same-party coworker shares. N = 37,200,614 positions for 30,965,666 unique workers. Analysis based on two-party exposure measures using imputed partisanship.

levels of segregation (10.2 pp, P < 0.001, 95% CI [0.091, 0.113]), while Millennials show the highest (12.2 pp, P < 0.001, 95% CI [0.109, 0.135]), with Gen X (11.6 pp, P < 0.001, 95% CI [0.104, 0.128]) and Gen Z (11.7 pp, P < 0.001, 95% CI [0.104, 0.131]) falling between these extremes. This suggests that younger cohorts who entered the labour market during periods of heightened political polarization may be more likely to sort into politically homogeneous workplaces.

Senior executive-level workers show the highest raw segregation (14.7 pp, P < 0.001, 95% CI [0.134, 0.161]). This may be because senior executives have the greatest power to select into firms with coworkers who share their partisanship or shape/select the partisanship of their subordinates. When examining job zones—which capture the skill, education and training requirements of occupations—we find a U-shaped pattern: jobs requiring little preparation (Zone 1: 12.1 pp, P < 0.001, 95% CI [0.111, 0.131]) and jobs requiring the most extensive preparation (Zone 5: 11.8 pp, P < 0.001, 95% CI [0.106, 0.130]) both exhibit high segregation, while middle-skill jobs show somewhat lower levels. However, after controlling for occupation, industry and geography, workers in jobs requiring extensive preparation (Zone 5) show the highest segregation (3.1 pp, P < 0.001, 95% CI [0.029, 0.033]), suggesting that highly skilled workers with substantial market power are particularly likely to sort into politically compatible workplaces.

Taken together, these heterogeneity patterns, paired with the preceding fixed-effect analysis, suggest that workplace political segregation is not simply a mechanical result of residential, industry or occupational sorting, but reflects active choices by workers who have both the motivation (political engagement) and opportunity (market power, flexibility) to select into politically compatible work environments.

Asymmetry in out-party exposure

The density plots in Fig. 1 establish that Democrats and Republicans work in systematically different environments: Republicans are substantially more likely to work alongside Republican coworkers than Democrats are. To examine the broader implications of this asymmetry for cross-party workplace contact, we analyse the distribution of both out-party and in-party exposure across the full range of worker experiences.

Democrats, on average, work in environments where 63% of their coworkers share their partisanship, while Republicans work in environments where only 44% of their coworkers share their partisanship. This translates to out-party exposure rates of 32% for Democrats and 50% for Republicans—a substantial 18 percentage point difference that reflects the fact that Democrats outnumber Republicans in the labour market, a feature of our sample and nationally representative estimates (see ‘Sample benchmarking’ in Methods).

However, focusing solely on average exposure obscures important variation in cross-party contact across the distribution of workers. Figure 6 presents a granular analysis of this variation by displaying both out-party and in-party exposure across percentiles of the exposure distribution, separately for Democratic and Republican workers. The heat maps reveal several important patterns that extend beyond simple averages.

First, the out-party exposure heat map (Fig. 6a) shows that even among Democrats in the highest deciles of Republican exposure—those most likely to work alongside Republicans—out-party contact remains relatively modest. By contrast, Republicans in the highest deciles of Democratic exposure experience substantially higher levels of cross-party contact, with some Republicans working in heavily

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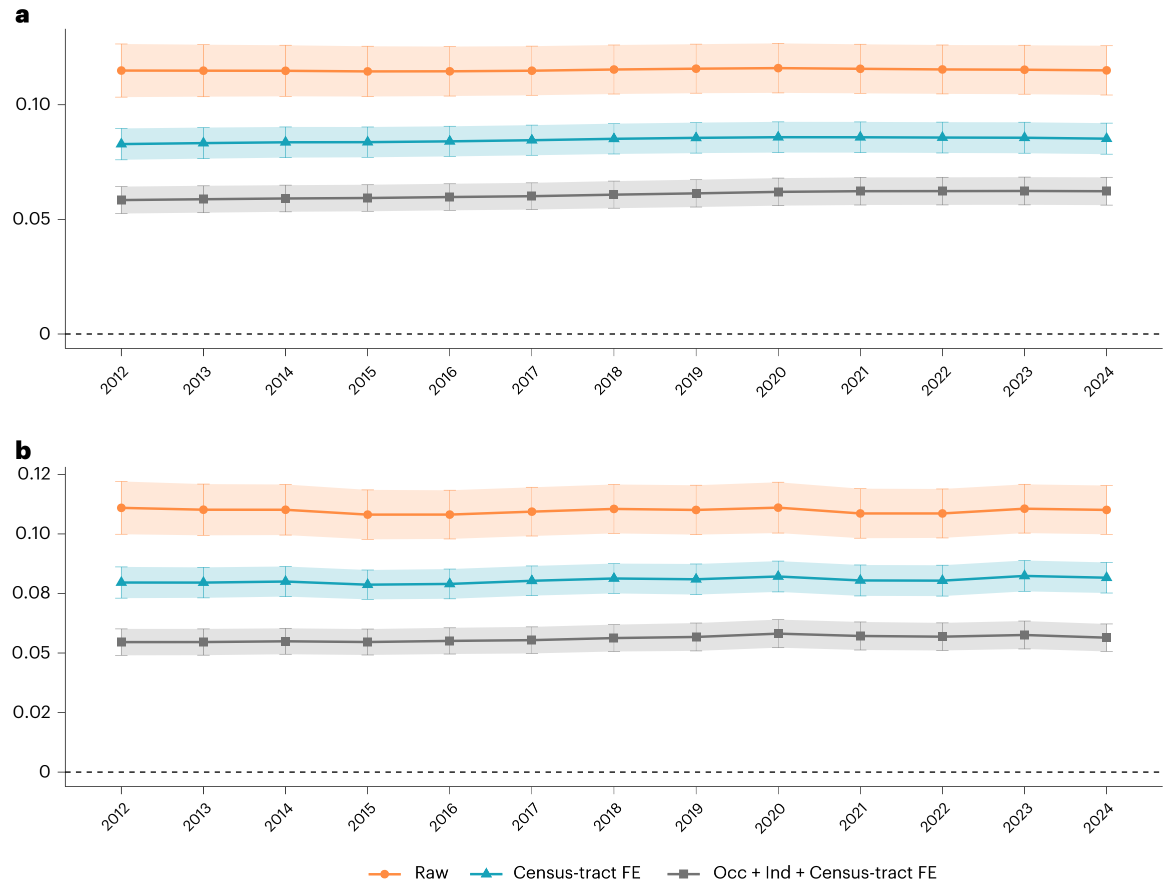

Fig. 7 | Longitudinal analysis. Coefficients from longitudinal regression analysis of Republican workplace segregation from 2012–2024. **a**, Results for the full sample of all position–years. **b**, Results restricted to workers experiencing employer changes. Three model specifications are shown: raw estimates with no fixed effects, census-tract FE models and occ-ind-census-tract FE models. Points represent OLS coefficient estimates; error bars represent 95% confidence intervals. The dependent variable (*y* axis) is the proportion of Republican coworkers in the MSA–employer cell, calculated using only Democrats and Republicans (imputed). Standard errors are clustered by MSA. Occ, occupation; Ind, industry. Complete regression results are reported in Supplementary Table 5.

Democratic environments. This asymmetry reflects the dual influence of Democratic workers' numerical advantage and workplace sorting patterns.

Second, the in-party exposure heat map (Fig. 6b) reveals the mirror image: Democrats consistently experience higher same-party contact across nearly all percentiles, while Republicans show greater variation in their in-party exposure. The most isolated Republicans (those in the lowest percentiles of Republican exposure) work in dramatically different environments than the most isolated Democrats, highlighting how political workplace segregation affects the two parties differently.

This distributional analysis demonstrates that political workplace segregation creates systematically different experiences for Democratic and Republican workers. While both parties exhibit workplace sorting, Republicans are more likely to find themselves in politically heterogeneous work environments, whereas Democrats are more likely to work primarily alongside other Democrats. These patterns have important implications for theories of workplace contact and political polarization, as they suggest that opportunities for cross-party interaction are distributed unequally across the political spectrum.

Temporal trends

Next, we examine temporal trends in workplace segregation by partisanship using the panel of position–years described in Methods. We re-estimate equations (2) and (5) with census-tract fixed effects, as well as the variant that includes occupation, industry and census-tract fixed effects. We plot the by-year estimates in Fig. 7. While the degree of overall political segregation increased modestly over the past decade, the relative stability of this estimate may reflect the fact that job changes occur infrequently. Accordingly, we repeat this analysis but restrict the sample to instances where workers start a new position, meaning they are either (1) entering the labour market for the first time or (2) switching employers. These results are reported in Fig. 7b. For the raw estimate, the degree of segregation remained essentially flat, declining slightly from 11.1 percentage points in 2012 ($P < 0.001$, 95% CI [0.100, 0.122]) to 11.0 percentage points in 2024 ($P < 0.001$, 95% CI [0.100, 0.120]). The census-tract fixed effects estimate shows a modest upward trend from 8.0 percentage points ($P < 0.001$, 95% CI [0.073, 0.086]) in 2012 to 8.2 percentage points ($P < 0.001$, 95% CI [0.075, 0.088]) in 2024, an increase of 0.2 percentage points over 12 years. The occupation-industry-census-tract fixed effects estimate shows a similar modest increase from 5.5 percentage points ($P < 0.001$, 95% CI [0.049, 0.060]) in 2012 to 5.6 percentage points ($P < 0.001$, 95% CI [0.051, 0.062]) in 2024, rising by 0.18 percentage points. Overall, the substantive magnitude of change is relatively small. One limitation of our data is that we cannot capture individuals who moved, changed registration status, or were not registered in 2024.

Discussion

Recent years have witnessed an outpouring of research examining the extent, origins and consequences of political segregation across various geographic and social domains[1,11,13]. Despite the workplace's potential for fostering prejudice-reducing, cross-partisan contact[20,21], very little is known regarding political segregation across workplaces. This paper addresses this gap by providing a large-scale measurement of the extent of political segregation in the US workplace. As we show here, the magnitude of political segregation is larger than analogous estimates of political residential segregation and similar in magnitude to workplace segregation based on gender. Moving beyond our main, 'raw' estimate, we present measures that include fixed effects for geography, industry and occupation. Our cross-metropolitan analysis (Fig. 2b) reveals that residential and workplace segregation co-vary across metropolitan areas, although substantial variation in workplace segregation patterns across areas remains. We show that even after accounting for geographic, industry and occupational political segregation, workplaces remain significantly sorted by partisanship. Importantly, these results are robust to reweighting our sample using BLS occupational data to address potential selection bias in our data as well as a wide range of alternative estimation and measurement decisions. These analyses provide confidence that our findings reflect genuine workplace sorting patterns rather than sampling artefacts.

Our findings illuminate the need for future research into the 'causes' and 'consequences' of workplace political segregation. First, regarding 'causes', political segregation may be driven by individual choices (or constraints) on where those from different political groups live as well as the occupations and industries in which they work. This does not mean that political segregation is merely incidental or that politics do not play a role. Partisans may make politically informed decisions about where to live and in which industry and/or occupation to seek employment based on their perception of the political makeup of these domains. For example, Democrats may seek out big-city jobs in education or media to work alongside other Democrats. Moreover, copartisan workers may make similar choices that are correlated with, but not necessarily driven by, partisanship. Research on political residential sorting finds that political homophily is largely driven not by explicit political bias but by the tendency of opposing partisans to prefer different neighbourhood amenities[48]. Similarly, Republicans and Democrats may hold different worldviews that lead them towards different occupations and industries. Political segregation by geography, industry and occupation may be mutually reinforcing. For instance, the tendencies of Democrats to live in more urban areas and Republicans to live in more rural regions may shape industry and occupation choices, which in turn may further influence or constrain the ability of these workers to relocate to other geographic regions.

Second, political segregation could arise during the hiring process as a result of employers' or job seekers' preferences for copartisan coworkers. Experimental evidence suggests that employers prefer to hire copartisans and discriminate against members of the opposing party[30,31]. While employers cannot legally discriminate on the basis of characteristics such as gender, race/ethnicity or age, political partisanship is not a protected category at the federal level or in most US states. Political discrimination in the hiring process need not necessarily be motivated by copartisan favouritism. Employers may prefer workers who match the majority party to promote a harmonious and collaborative workplace[40]. Job seekers may also show greater propensity to seek work at employers where they believe most coworkers will share their partisanship. This may be driven by job seekers' desire to work alongside copartisans and/or concern that they will face discrimination at the hands of outpartisans. As companies increasingly take public stances on salient political issues, it may become easier for job seekers to infer and thus select into applicant pools based on the political composition of coworkers. Our heterogeneity analyses provide correlational evidence consistent with this mechanism. Workers who are more politically engaged (and thus plausibly more likely to care about their coworkers' partisanship) and who hold more labour market power (and are thus better able to realize these preferences) are more politically segregated.

Third, partisan segregation could be driven by processes that occur after job seekers join employers. Workers in the political minority within their employer may either come to adopt the majority partisanship over time (socialization) or they may be more likely to leave (differential attrition). Previous literature suggests that both effects can and do occur. Political socialization as a result of peer influence has been documented within residential neighbourhoods[49], colleges[50] and explicitly political workplaces (that is, politicians' offices)[51], whereas differential attrition by political 'misfits' has been shown in several workplace contexts[52,53]. However, work in progress[35] suggests that this socialization may be modest and limited to non-partisans.

Before turning to consequences, we note several measurement limitations. First, our worker profile data are drawn predominantly from online professional platforms, which overrepresent white-collar and college-educated workers relative to the broader US labour force. If political sorting is stronger among knowledge workers—who face fewer geographic constraints and have more employer options—our segregation estimates may represent an upper bound on workforce-wide levels. Second, we successfully match 31.8% of voter file records to employer profiles; easier-to-match workers (those with distinctive names or detailed online profiles) may differ systematically in their political behaviour or occupational distribution. Our BLS-reweighted robustness checks address observable dimensions of this selection, but selection on unobservables cannot be ruled out. Third, employer-level attributes, including firm names, NAICS industry codes and O*NET occupational classifications, rely on automated entity resolution, which we validated manually for the largest employers but which may contain errors for subsidiaries, post-merger entities and multibrand corporate structures. Fourth, our partisanship measure has inherent limitations: not all states record party registration, registration does not always correspond to self-identification (particularly in closed-primary states where strategic registration may occur), and the Democrat–Republican binary is coarse, obscuring within-party heterogeneity in policy preferences, issue salience and intensity of partisan attachment. Our robustness checks using alternative partisanship operationalizations suggest that these limitations do not drive our main results, but they constrain the precision with which we can characterize workplace ideological composition. Fifth, our analysis is limited to metropolitan statistical areas, excluding ~12% of the US workforce in non-metropolitan areas where partisan composition and labour market structure may differ.

A broader interpretive constraint bears emphasis. Our analyses are primarily cross-sectional and descriptive: we document the extent of political segregation but cannot distinguish among the mechanisms that produce it—whether partisan sorting during job search, employer discrimination in hiring, workplace socialization, or differential attrition of political minorities. Each could independently generate the patterns we observe, and they probably operate simultaneously. Our temporal analysis of new hires provides suggestive evidence consistent with sorting, but this evidence does not resolve the identification challenge, as trends in new-hire composition are themselves consistent with multiple mechanisms operating in parallel.

Beyond elucidating the drivers of workplace political sorting, future research can investigate its 'consequences'. First, research is urgently needed to understand the relationship between workplace political segregation and affective polarization. The workplace is perhaps uniquely well suited to foster the type of intergroup contact that can potentially reduce prejudice[20,21]. Given growing concerns surrounding sharp rises in political discord and affective polarization in the United States[15], understanding whether the workplace serves as a locus of prejudice-reducing cross-partisan contact is imperative. While we have emphasized political segregation in presenting our results, the data we present also point to meaningful levels of outpartisan exposure, especially among Republicans. On this front, a critical question left unanswered by our study is whether workplaces with less political segregation are in fact more likely to foster meaningful cross-partisan contact and whether these interactions decrease affective polarization. To address this question, subsequent enquiry could use surveys to identify conversation networks as well as the extent and nature of workplace conversations around politics.

Second, workplace political segregation may be linked to the growing tendency of businesses to take public positions on controversial social issues. These stances have appeared puzzling, given that the benefits from appealing to like-minded stakeholders are often outweighed by the negative reactions from opposing stakeholders[28,54]. One suggestion is that employees may drive firms' engagement with controversial political issues[38]. Research mapping political segregation to corporate sociopolitical activism may be able to elucidate whether corporate sociopolitical positioning is primarily a strategic response to workers' demands as opposed to a unilateral expression of top managers' beliefs or values[55,56].

Finally, workplace political segregation may affect firm performance. On one hand, political homogeneity arising from political segregation might enhance firm performance. Employees who feel politically misaligned with upper management or their peers may be less economically productive[27,29,57,58] and more likely to exit the firm[52,53]. Moreover, if workers value politically homogeneous workplaces where they are in the majority, they may even accept a lower wage for jobs in these workplaces[29]. On the other hand, political segregation might erode firm performance. Studies of labour force segregation based on gender have shown that resulting (mis)allocations of talent across industries and occupations can hinder economic growth[59]. To the extent that political sorting may also result in a sub-optimal allocation of talent[60], political segregation may have important economic consequences. Understanding the relationship between political segregation and firm productivity, innovation and profitability represents an important frontier for future scholarship.

Methods

Ethical considerations

This study was reviewed by the Health Sciences and Behavioral Sciences Institutional Review Board (IRB-HSBS) at the University of Michigan and determined to be 'Not regulated' under the University's policy for research using publicly available datasets, in accordance with federal regulations for human participants research (45 CFR Part 46) [HUM00258211, determined 22/7/2024]. The study linked administrative data from two commercial sources: L2, Inc. (voter registration records) and Revelio Labs (employment profiles derived from LinkedIn), using names and metropolitan area information. While the underlying source information (state voter registration records and publicly available LinkedIn profiles) is publicly accessible, the compiled and processed datasets used in this study are proprietary commercial products. Data were collected and processed in compliance with applicable privacy regulations and University of Michigan data security policies. All analyses were conducted on secure university servers with dual-factor authentication. To minimize identifiability risks from the merged datasets, we do not release individual-level data, report only aggregate analytical results and have anonymized all released data samples, including perturbing key variables with noise and hashing all identifiers, in compliance with commercial licensing agreements. The replication dataset contains no direct personal identifiers and cannot be used to re-identify specific individuals.

Preregistration

This study was not preregistered.

Overview

We estimated political segregation in the workplace by linking state voter registration files with a database of online worker profiles. This approach meaningfully improves on previous studies of workplace partisanship in the United States in terms of the number and representativeness of workers it captures. Most existing research on firm-level political partisanship (and ideology) relies on data from legally mandated public disclosure of individual political donations by the FEC. Because these public campaign finance disclosures contain names, addresses, job titles and employer information, researchers have used data on partisan donations to measure firm-level partisanship and/or as a proxy for political ideology (for example, refs. 61,62). Many previous studies leverage the extensive Database on Ideology, Money in Politics, and Elections (DIME) data, developed by ref. 36, which not only captures this donation data, but also precisely characterizes donors' ideology based on the ideology of the candidates to whom they donate. (Political partisanship and ideology are theoretically distinct concepts, although they are highly correlated in the modern US political landscape. One major strength of the DIME data is that it can make fine-grained ideological distinctions between more moderate and more extreme members of the same party. As we discuss, our measure of partisanship greatly expands coverage but at the cost of a coarser measurement.)

The major trade-off for this measurement precision is narrow and skewed coverage. The vast majority of Americans do not donate to political campaigns, let alone in amounts large enough that historically required disclosure. As a result, donations-based measures cover a narrow and demographically unrepresentative slice of the US population, one that is not only far more politically engaged, but also far whiter, wealthier and older than the US population as a whole[41,42]. Thus, while DIME and other campaign finance-based datasets may provide good coverage for highly educated, wealthy and politically engaged professions such as lawyers[63] and doctors[64], they are less representative when looking at employees or companies in general[65].

In addition to questions about representativeness, linking FEC data with companies can also be technically challenging. FEC data are based on reports filed by political campaigns or other political organizations, which must make a good faith effort to collect accurate information from their donors. However, unlike the voter file, where registered voters must provide their exact legal name to vote, there is nothing preventing donors from providing variants or nicknames. Company names are also not standardized, nor can corporate parents easily be linked with subsidiaries[62]. Donors may choose to withhold information, leading donation filings to read only that information has been 'requested'. Furthermore, donors may also give technically accurate but misleading information about their employment, such as listing a position in an industry trade association rather than their primary employer—a tactic that is especially common among firms facing reputational challenges[66].

Given both the limited coverage and technical challenges involved in linking FEC disclosures with companies, previous efforts to link individual workers' partisanship with their employer have been orders of magnitude smaller than the data we present here. For instance, ref. 62 deployed extensive code to standardize and parse federal campaign contributions from 2003 to 2016 and was able to identify only 85,109 individuals across 874 political action committees (PACs) associated with publicly traded companies (see also ref. 67). Another analysis[40] estimated that 60% of US companies have no donors that appear in DIME.

For some studies, such as those that focus on the ideology of the 'upper echelons' within a firm or those which are primarily concerned with PAC contributions, these campaign finance measures may be appropriate. When it comes to capturing the general employee population, we believe our approach provides more comprehensive coverage compared with existing donation-based measures. We discuss the representativeness of our sample in more detail in 'Sample benchmarking' below.

L2 voter file

Our first data source is the national voter file. In the United States, individuals' vote choices are secret, but whether they register or turn out to vote is a matter of public record. Voters' registration and turnout history, combined with demographic and contact information, are widely used by political campaigns, commercial data vendors and marketers, and, increasingly, by academic researchers[68]. In many states, these data can be accessed by contacting individual state and/or local officials, but researchers more typically rely on data vendors which aggregate, clean and standardize the raw records received from election officials. Voter file data have been widely used in political science, economics and other fields, often by merging these data with other datasets to study topics such as voter turnout[69,70], as well as the political makeup of various professions and groups, including board members[71], college students[50], government bureaucrats[58], patent examiners[72], physicians[73], the police[74], religious leaders[9] and spouses[75]. Our voter file comes from the non-partisan vendor L2 Data, extracted and processed in November 2024, and contains information on ~185 million registered voters in the United States. Estimates from the 2022 Cooperative Election Study suggest that ~73% of the overall US adult population are registered to vote; among those currently employed, it is 76%.

Below we describe the variables we used from the L2 voter file:

Partisanship. Unlike many other countries (for example, Brazil; see ref. 31), political parties in the United States do not maintain nationwide member rolls. Instead, L2's information on partisan identification comes from a variety of sources, which vary by state[69]. States fall into three categories (see https://www.l2-data.com/wp-content/uploads/2024/09/State-by-State-Partisanship-Party.pdf):

- 30 states (and Washington, D.C.) register voters by party.
- 11 states do not formally register voters by partisanship, but do record whether voters chose to vote in either party's primary and classify voters in this manner. (These states are Georgia, Illinois, Indiana, Michigan, Mississippi, Ohio, South Carolina, Tennessee, Texas, Virginia and Washington.)
- 9 states provide no information on partisan registration, and L2 models partisan affiliations based on ethnicity, geography and other data. (These states are Alabama, Arkansas, Hawaii, Minnesota, Missouri, Montana, North Dakota, Vermont and Wisconsin.)

A report by Pew Research that compared modelled data on commercial voter files with self-reported survey responses found that modelled partisanship is correct in a majority of cases[76].

Partisanship imputation. In our main analysis, we imputed the partisanship of workers who are registered as independents, non-partisans or with third parties. Among registered voters in our sample, about one-third were not registered as Democrats or Republicans. However, party registration potentially understates de-facto partisan segregation, as many independents have clear partisan leanings[44–46]. Following the methodology of ref. 1, we imputed partisan 'lean' for non-partisans using a multistep approach, detailed in Extended Data Fig. 1. First, we began with the partisan estimates provided by L2, as described previously. Second, for voters who were not classified as either Republican or Democrat, we identified whether they have a previous history of voting in either the Democratic or Republican political primary from 2012–2022, and assigned them to the party in whose primary they most recently voted. Third, we assigned members of third parties as 'leaning' towards the party that most closely matches the ideology of their preferred third party. For example, the Green, Socialist, and Working Family Party are mapped to the Democratic party, while the American Independent, Libertarian, and Reform parties are mapped to the Republican party. We adopted the party classifications from ref. 1.

Approximately 10–30% of voters remain unclassified at this point, depending on the state. To model the partisan lean of these voters, we relied on a Bayesian approach[1] which incorporates fine-grained information about the local partisan geography as well as demographic modelling. First, we calculated a Bayesian geographic prior that is based on precinct-level electoral returns. As precinct-level returns were not yet available in all states for 2024, we used 2020 returns. We attempted to assign voters to a precinct using spatial joins with the `R` sf package; in the rare cases where we were unable to do so, we used county-level returns. This approach largely mirrors that done by ref. 1. We departed slightly from their procedure by not incorporating the number of registered Republicans and Democrats and instead relying only on vote totals for the Republican and Democratic candidates (based on personal communication with one of the authors, we confirmed that this approach delivers comparable results).

Second, we incorporated information about voters' demographics. We began with a large-scale, high-quality survey (the 2020 Cooperative Election Study) and used these data to estimate the conditional probabilities that an individual falls within a given demographically defined stratum (defined by age bracket × race/ethnicity × gender × voter registration status) conditional on them identifying as a Democrat, Republican or true (non-leaner) independent. (In cases where race/ethnicity data were missing in L2, we used only age, gender and voter registration status.)

We then used Bayes' formula to incorporate the demographic information and the geographic prior to estimate the probability that each 'independent' voter actually leans Democratic or Republican. We assigned each voter as 'leaning' towards the partisan label with the highest probability (that is, Republican, Democratic or Independent). (In the highly unlikely event that all probabilities are equal, we defaulted to independent.) This imputation procedure classified the majority of initially unaffiliated voters as Democratic or Republican leaners, with only a small fraction remaining as true independents.

Gender. Gender classification in L2 combines direct collection and probabilistic imputation. Twenty-seven states directly include gender in their voter registration files, while 22 states and Washington, D.C., do not collect this information. (The states that include gender are Alabama, Alaska, California, Colorado, Connecticut, Florida, Georgia, Idaho, Illinois, Indiana, Iowa, Kansas, Kentucky, Louisiana, Maryland, Massachusetts, Michigan, New York, North Carolina, Pennsylvania, Rhode Island, South Carolina, Tennessee, Texas, Virginia, Washington and West Virginia.) Of these, California includes gender in voter files, but the field is missing for a majority of registered voters.

For voters where gender is not directly provided, L2 implements probabilistic gender imputation based on first name frequency distributions. This approach uses administrative records and Census data to calculate the probability that individuals with specific first names are male or female, then assigns gender based on the higher probability. When first names are ambiguous (roughly equal gender probabilities), L2 incorporates middle name information to improve classification accuracy (based on personal email exchanges with L2).

Geographic controls. To match datasets and control for residential sorting, we assigned users to geographic areas. We matched on MSA, which is assigned by Revelio based on the user's reported workplace. We assigned L2 users to MSAs based on the HUD User (https://www.huduser.gov/) crosswalks.

We used census-tract assignments, derived from L2's geocoding home addresses to census-tract boundaries, to implement geographic controls. Census tracts represent small, relatively homogeneous areas designed to contain 1,200–8,000 residents with similar demographic and socioeconomic characteristics. This allowed us to compare workers who live in the same neighbourhood but work for different employers.

Political participation. All political participation variables came from L2. To measure registered voters' degree of political participation, we used L2 data on voters' turnout in general and primary elections.

We also included a binary indicator for whether the individual was a political donor, based on whether they gave at least one donation to a political campaign, as captured under FEC donation requirements.

Worker profiles from Revelio Labs

Previous work in other national contexts has leveraged administrative tax or pension data to study workplace partisanship (for example, ref. 31). In the United States, federal administrative data are not publicly available and are made available to researchers only under certain narrow circumstances. Statutory guidelines for the use of personally identifiable tax or social security microdata for research purposes probably preclude their use for studies that focus on political partisanship. For example, researcher data access often relies on sampling or the use of synthetic and/or coarsened anonymized data, neither of which would be suitable for our purposes. While there have been notable cases where researchers have gained access to Internal Revenue Service (IRS) data to study less politically charged topics such as economic mobility (for example, ref. 77), these have only been for subsets (that is, individual birth cohorts) rather than the entire US workforce, as well as on research topics that are comparatively less politically sensitive relative to partisan politics. Because these data are not available, we relied on data on job positions from Revelio Labs. Revelio is a workforce intelligence company that uses proprietary technology to compile individual employment records based on online professional profiles (that is, LinkedIn), as well as job postings. Revelio data have been used in a variety of studies in management, accounting and other disciplines to study topics including ESG reporting and workplace diversity[78–82], the drivers and consequences of employee turnover[83–85], corporate culture[86], and the influence of individual work history and life experiences on business outcomes[87,88]. The Revelio data we used contain information on individual job positions, including the date on which they started and ended. Our particular Revelio dataset was captured in April 2025. After excluding records with missing data fields (MSA or employer), our Revelio data available for matching covered ~129 million positions, held by 103 million unique workers.

Below we detail each of the variables we used from this dataset.

Names and personal information. Revelio extracts names and professional information directly from LinkedIn profiles as they appear on the platform. However, we recognize that LinkedIn names may differ from legal names used in voter registration due to professional naming preferences, nicknames, maiden names or cultural naming conventions. To address this challenge, we implemented name cleaning using the 'nominally' Python library, which standardizes name formats, handles common variations and parses complex name structures. This preprocessing substantially improves matching accuracy compared to using raw LinkedIn name data.

Employer information. Revelio relies on LinkedIn's entity relationships and implements name disambiguation and entity resolution to address the fundamental challenge that the same company may appear with multiple name variations across LinkedIn profiles (for example, 'BofA,' 'Bank of America,' 'BoA,' 'Bank of America Corp'). Using machine learning algorithms trained on business registration records, Securities and Exchange Commission (SEC) filings and other commercial databases, Revelio groups variant employer names into standardized entities.

Industry classification. Revelio classifies employers into NAICS codes at the sector, subsector and detailed industry levels. This process combines automated matching of employer names against commercial business databases, with machine learning models used to predict industry codes based on company names, descriptions and employee job titles. The resulting classifications provide comprehensive industry coverage but represent another area of proprietary processing that we treat as authoritative.

Occupation classification. Job titles from LinkedIn are mapped to standardized O*NET-SOC occupation codes using Revelio's proprietary natural language processing pipeline. This system analyses job titles, job descriptions (when available) and contextual information from job postings to predict the most appropriate occupation code. The model incorporates information about job responsibilities, required skills and industry context to improve classification accuracy.

Seniority and career progression. Revelio estimates career seniority using a 7-category classification ranging from entry-level positions (interns, trainees) to senior executive roles (C-suite). This classification combines multiple information sources: (1) current job title analysis using natural language processing models trained to recognize seniority indicators, (2) individual employment history showing career progression patterns and (3) estimated age based on years of employment history. The age estimation itself represents a substantial methodological challenge, as LinkedIn does not provide birth dates; based on personal communication with Revelio Labs, Revelio infers age from career duration and typical education-to-work transition patterns.

Ensemble matching strategy

Merging datasets is not a new challenge in social science research[89]. However, this particular application presents unique challenges. First, the Revelio data are sourced from online employment profiles, rather than from government or administrative sources. As a result, we lack the individual identifiers—for example, dates of birth, social security numbers or home addresses—that are commonly used to link datasets. Furthermore, individuals may use an entirely separate name professionally (for example, a nickname, middle name or maiden name) that does not correspond to the legal name they use when registering to vote. While social scientists have developed methods for linking administrative datasets, these are typically designed to address partially missing data, data entry errors (for example, typos) or inconsistencies (for example, inconsistent usage of abbreviations), rather than cases where bridging data fields are almost entirely absent. A second related challenge is the sheer size of the data. Limiting to observations within MSAs, the L2 voter file contains ~177 million voter records and the Revelio data contain ~103 million distinct users, meaning that a brute force attempt to merge the two datasets would involve ~1.8×10^{16} (18 quadrillion) pairwise comparisons.

To address these challenges, we implemented an ensemble matching approach that combines two complementary methodologies: probabilistic record linkage using the Fellegi–Sunter framework and LLM-based semantic matching. Our data snapshots combined the most recent available data from both sources. In L2, we used voter file extracts processed in November 2024, while in Revelio we used employment data captured in April 2025. (As a result, this means that we only matched workers who were currently registered to vote as of the 2024 L2 extract. Most notably, individuals who had work history but who died before the L2 extract date are probably removed from the voter rolls and thus not available to be matched.)

Data preprocessing. We began by partitioning both the Revelio and L2 records by MSA. We excluded records where the work or home address are located outside of an MSA or where location data are missing. According to the 2021 American Community Survey (ACS) DP03 table, nearly 88% of Americans in the labour force live within an MSA. This is done for both practical reasons (to make the computation feasible) and theoretical reasons—MSAs are constructed by the Census Bureau to capture commuting patterns, so it is reasonable to limit our matches to cases where the work location (as reported in Revelio) and home address (as reported in L2) are within the same MSA. For MSAs that span multiple states, we attempted to match job positions with voter registrations for all states associated with that MSA (for example, we would attempt to match job positions located in the

Philadelphia–Camden–Wilmington, PA-NJ-DE-MD MSA with voters registered in the MSA in PA, NJ, DE and MD).

Probabilistic matching (Splink). Our first matching approach used Splink v.3.9.8 with a DuckDB backend to implement probabilistic record linkage following the Fellegi–Sunter framework[33]. This approach estimates the probability that two records refer to the same individual based on the agreement patterns across multiple comparison variables. We used string comparison functions, including exact matches, Damerau–Levenshtein distances (threshold ≤1) and Jaro–Winkler similarity scores (thresholds ≥0.8 and ≥0.9) for first and last names. Additional comparisons included birth dates (exact year, ±1 year and ±10 year matches), middle names (using name comparison functions) and exact gender matches. All comparisons incorporated term frequency adjustments to account for the relative commonness of names and demographic characteristics. The model uses blocking on last name to reduce computational complexity, with expectation-maximization to estimate match probabilities. For large MSAs, we implemented name binning strategies (grouping by first letters of last names) to manage memory requirements during model training.

LLM-based matching (fuzzylink). Our second approach used the fuzzylink R package[34] to perform embedding-based semantic matching powered by large language models. This method uses GPT-4o-mini-2024-07-18 for match classification and OpenAI's text-embedding-3-large model (256 dimensions) to generate semantic embeddings of names. The approach blocks on gender and last name, then uses embedding similarity to identify potential matches. We provided specific instructions to the LLM to ignore professional titles, suffixes, case differences, punctuation, accents, name order variations and minor misspellings while focusing on whether names likely refer to the same person. This method is particularly effective at capturing matches missed by traditional string-based approaches, such as those involving nicknames, initials or alternative name presentations common in professional contexts.

Ensemble integration. We combined the results from both matching approaches using an ensemble strategy that maximizes coverage while maintaining match quality. Our algorithm prioritized matches with the highest match probability by removing duplicate Revelio user IDs (keeping the highest match probability), then removing duplicate voter IDs (again keeping the highest match probability). For ties, we prioritized records with quality indicators (for example, Revelio's 'is_bad_user' flag). If there are still ties, we randomly selected which match to include.

Our ensemble approach yielded ~45.3 million unique matched workers representing ~31.8% of Revelio users and 18.9% of registered voters, constituting the largest matched employer–voter dataset ever constructed.

Sample attrition and waterfall analysis. The construction of our analytical dataset involved substantial but methodologically necessary attrition at each processing stage. We documented this process through waterfall charts (presented in Supplementary Figs. 11 and 12) that track observation losses from initial data sources to final analytical sample.

Beginning with 102.6 million unique workers in Revelio's US MSA workforce and 176.7 million registered voters in L2, our ensemble matching process successfully linked 44.2% of Revelio workers and 27.2% of L2 voters, resulting in the loss of 57.3 million Revelio workers and 128.6 million L2 voters. This substantial but expected attrition reflects the conservative nature of our matching approach, which prioritizes precision over recall to ensure high match quality. The differential retention rates between datasets reflect their distinct population coverage: Revelio captures employed workers with LinkedIn profiles, while L2 encompasses all registered voters regardless of employment status or online presence.

Subsequent filtering steps created our 2024 analytical cross-section. We applied a number of restrictions, including: (1) requiring non-missing values for partisan exposure measures, imputed partisanship scores and demographic variables (gender, age); (2) restricting to workplaces with at least two employees to enable meaningful coworker exposure calculations; (3) requiring complete workplace characteristics including valid industry (NAICS) and occupation (O*NET-SOC) classifications; (4) restricting to workers classified as Democrats or Republicans using our imputed partisanship methodology, excluding independents and third-party registrants; and (5) requiring valid geocoded residential addresses for census-tract assignment to enable geographic fixed effects.

Our final analytical sample comprised 31 million unique individuals from the Revelio perspective and 34.8 million from the L2 perspective, representing 30.2% and 18.6% retention, respectively. The difference between these final counts reflects the fact that L2 identifiers are state specific. Therefore, a person who registers to vote in two states over their lifetime will receive two L2 identifiers.

Summary statistics

Summary statistics for our working dataset are presented in Extended Data Table 2. Our final analytical sample comprised 37.2 million observations representing 31 million unique individuals working across 366 metropolitan statistical areas. The sample spanned 1,013 industries and 383 occupations. Throughout the paper, we refer to employer–MSA combinations as 'workplaces'. Consistent with the highly skewed distribution of employees across employers in the United States, most employers had a small number of workers, although a small number of employers were very large, with workplace sizes averaging 904 employees.

After applying our partisanship imputation methodology to assign probable Democratic or Republican affiliation to all workers, our analytical sample comprised 59.1% Democrats and 40.9% Republicans. The sample is 53.5% female and 39.1% non-white, with a mean age of 43 years. Political engagement was high, with 81.6% having voted in the 2020 general election and 3.5% having made federal campaign contributions. While the proportion of Democrats appeared larger than estimates of the share of the population who identify as Democrats, it is consistent with large-scale survey data that measure partisan 'registration' (as opposed to self-identification) among employed individuals. We discuss this at greater length in the 'Sample benchmarking' section, below.

Our main analysis reported in the Results section used a snapshot of the most current workforce identifiable using our data. This sample comprised positions that were reported as active at any point from the start of 2024 through our Revelio sample's end date of April 2025. In instances where a given worker had multiple positions within the same firm during this period, we prioritized the position that was ongoing or with the latest end date. This avoids double counting a worker's experience in a given firm.

We also used our merged data to create an unbalanced panel of positions spanning 2012 to 2024. We did so by considering all positions active at any point between 2012 and 2024, inclusive (not just those in active employment as of 2024), held by workers we were able to match to the November 2024 L2 state-level voter files. We then used the start and end dates for each position to define unique position–year dyads. Here, units of analysis are position–years, rather than simply positions, such that the same position appears separately in each year in which it was active at any time. Supplementary Fig. 4 charts the number of observations by year. The panel dataset statistics are similar to those for the 2024 cross-sectional data. Since our main objective is to estimate the most current degree of segregation in the United States labour market, we conducted our main benchmarking analysis and segregation estimates with respect to the 2024/2025 merged data. We

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used this panel of workers in the analysis described in the 'Temporal trends' section above.

Sample benchmarking

We benchmarked our merged sample against external population data to assess representativeness and identify potential sources of bias. This benchmarking is crucial for understanding the generalizability of our results and informs our weighting methodology. We conducted three types of comparison: demographic representativeness using the Cooperative Election Study (CES), industry coverage using the ACS, and occupational coverage using BLS data.

Political and demographic representativeness. Because our sample drew from registered voters with LinkedIn profiles, we faced potential selection bias along both political and demographic dimensions. To assess political representativeness, we compared our sample to the 2023 Cooperative Election Study, a large-scale, nationally representative survey. Since our measure of partisanship relied on official party registration rather than self-reported partisanship, we benchmarked against the population of registered voters in the CES who report working full- or part-time.

Extended Data Table 1 shows that our merged sample closely matched the population of working registered voters on key political dimensions. Among registered voters, our sample contained 41% Democrats compared to 43% in the CES, and 29% Republicans compared to 32% in the CES. The share of independents and third-party registrants (30% versus 25%) was also comparable. This close alignment provides confidence that our findings reflect genuine workplace sorting patterns rather than systematic political bias in our data sources.

Demographically, our sample showed some differences from population benchmarks. While our racial composition (61% white, 39% non-white) was reasonably representative compared to working registered voters (72% white, 28% non-white), it somewhat overrepresented non-white workers. Gender representation in our sample (53% women, 47% men) aligned well with the broader working population, although it slightly overrepresented women compared to registered voters specifically.

Industry coverage. Our reliance on LinkedIn profiles created systematic overrepresentation of white-collar industries and underrepresentation of blue-collar sectors. Extended Data Table 3 compares our sample's industry composition to 2023 American Community Survey data. Professional sectors where LinkedIn usage is common—financial activities (+8 pp), information technology (+6 pp) and manufacturing (+4 pp)—were substantially overrepresented. Conversely, industries where online professional profiles are less common showed noteworthy underrepresentation: construction (−5 pp), leisure and hospitality (−5 pp), and retail trade (−3 pp).

Occupational skill distribution. The industry patterns suggest that our sample was skewed towards higher-skill occupations. Extended Data Table 4 shows coverage across O*NET job zones, which measure the education, experience and training requirements for different occupations. Our sample dramatically overrepresented high-skill positions: Zone 4 occupations (requiring considerable preparation) comprised 41% of our sample versus 25% of the population (+16 pp), and Zone 5 occupations (requiring extensive preparation) represented 14% versus 5% of the population (+9 pp). Conversely, we substantially underrepresented lower-skill positions, with Zone 1 and Zone 2 occupations showing deficits of 5 and 21 pp, respectively.

These benchmarking results reveal systematic selection bias that could affect our segregation estimates. The overrepresentation of high-skill, white-collar occupations might artificially inflate political segregation estimates if such workers have more opportunity to sort into politically similar workplaces.

However, our political benchmarking provides reassurance that the most critical dimension for our analysis—partisan representativeness—is well preserved. The close alignment between our sample and population benchmarks on party registration suggests that while some occupational bias exists, it does not systematically favour one party over another in ways that would fundamentally undermine our segregation estimates.

BLS weighting methodology

To address the systematic occupational bias identified in our benchmarking analysis, we implemented a weighting scheme using BLS-OEWS data. Our sample might be systematically unrepresentative due to differential rates of: (1) voter registration across demographic groups, (2) LinkedIn profile availability and usage patterns and (3) matching success in our ensemble approach. Most notably, professional occupations appeared overrepresented relative to blue-collar occupations, potentially biasing our segregation estimates.

Our weighting methodology proceeded in several steps. First, we mapped O*NET occupation codes from our dataset to 2-digit SOC major occupation groups used by BLS, enabling us to link our observations to official employment statistics. Second, we standardized metropolitan area names to match BLS area definitions used in the OEWS data, accounting for variations in MSA naming conventions between our data sources.

Third, we calculated the observed distribution of workers across occupation–MSA combinations in our analytical sample, creating empirical proportions for each SOC major group within each metropolitan area. Fourth, we extracted benchmark employment distributions from the 2024 BLS-OEWS data, using metropolitan-area-specific employment counts where available and falling back to national proportions for areas not covered in the metro OEWS data.

Finally, we calculated sampling weights as the ratio of BLS benchmark proportions to our observed sample proportions for each occupation–MSA combination. Formally, for occupation o in metropolitan area m, the weight is defined as:

$$w_{o,m} = \frac{\text{BLS\_Proportion}_{o,m}}{\text{Sample\_Proportion}_{o,m}} = \frac{\frac{\text{BLS\_Employment}_{o,m}}{\sum_{o,m}\text{BLS\_Employment}_{o,m}}}{\frac{\text{LinkedIn\_Count}_{o,m}}{\sum_{o,m}\text{LinkedIn\_Count}_{o,m}}} \tag{6}$$

where $\text{BLS\_Employment}_{o,m}$ represents the official employment count from BLS-OEWS data for occupation o in MSA m, and $\text{LinkedIn\_Count}_{o,m}$ represents the number of matched workers in our sample for the same occupation–MSA combination. These weights effectively upweighted underrepresented occupation–MSA cells and downweighted overrepresented cells, adjusting our sample to match the national occupational distribution.

Use of AI tools

During the preparation of this work, the authors used Claude (Anthropic) for assistance with writing scripts, drafting and editing. The authors reviewed and edited all AI-generated output and take full responsibility for the content of this publication.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The Revelio and L2 datasets used in this paper are commercial datasets whose license terms prohibit disclosure. Datasets can be licensed for academic use by contacting Revelio Labs (reveliolabs.com) and L2 (l2-data.com). Interested researchers should contact these vendors directly to negotiate academic licensing terms. The authors are available to advise on the data access process. To facilitate transparency into the analytical methods used in this paper, we provide a partial random sample of observations that have been anonymized to protect individual privacy and comply with licensing agreements. Key variables have been perturbed with noise to further safeguard individual identities, and all identifiers have been hashed. This partial sample is available in Harvard Dataverse at https://doi.org/10.7910/DVN/1K7OUZ (ref. 90).

Code availability

We provide the R code used to produce the figures and tables in the main body of this article to demonstrate the methodology used throughout this paper. Combined with the sample data provided, this code can be used to demonstrate and evaluate the methods used in this paper, provide transparency into the research process, and allow researchers to understand and potentially adapt the methods used here for their own work. However, the figures and tables produced using this code and the sample data will not replicate those that appear in the paper. Code is available in Harvard Dataverse at https://doi.org/10.7910/DVN/1K7OUZ (ref. 90).

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Author contributions

All authors contributed to conceptualization, methodology, data curation and writing (original draft writing, review and editing). J.F. and M.K. performed the formal analysis, developed the software and created the visualizations. R.H. acquired resources and funding. J.F. also acquired funding. All authors supervised the project and contributed to project administration.

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Competing interests

The authors declare no competing interests.

Additional information

Extended data is available for this paper at https://doi.org/10.1038/s41562-026-02501-9.

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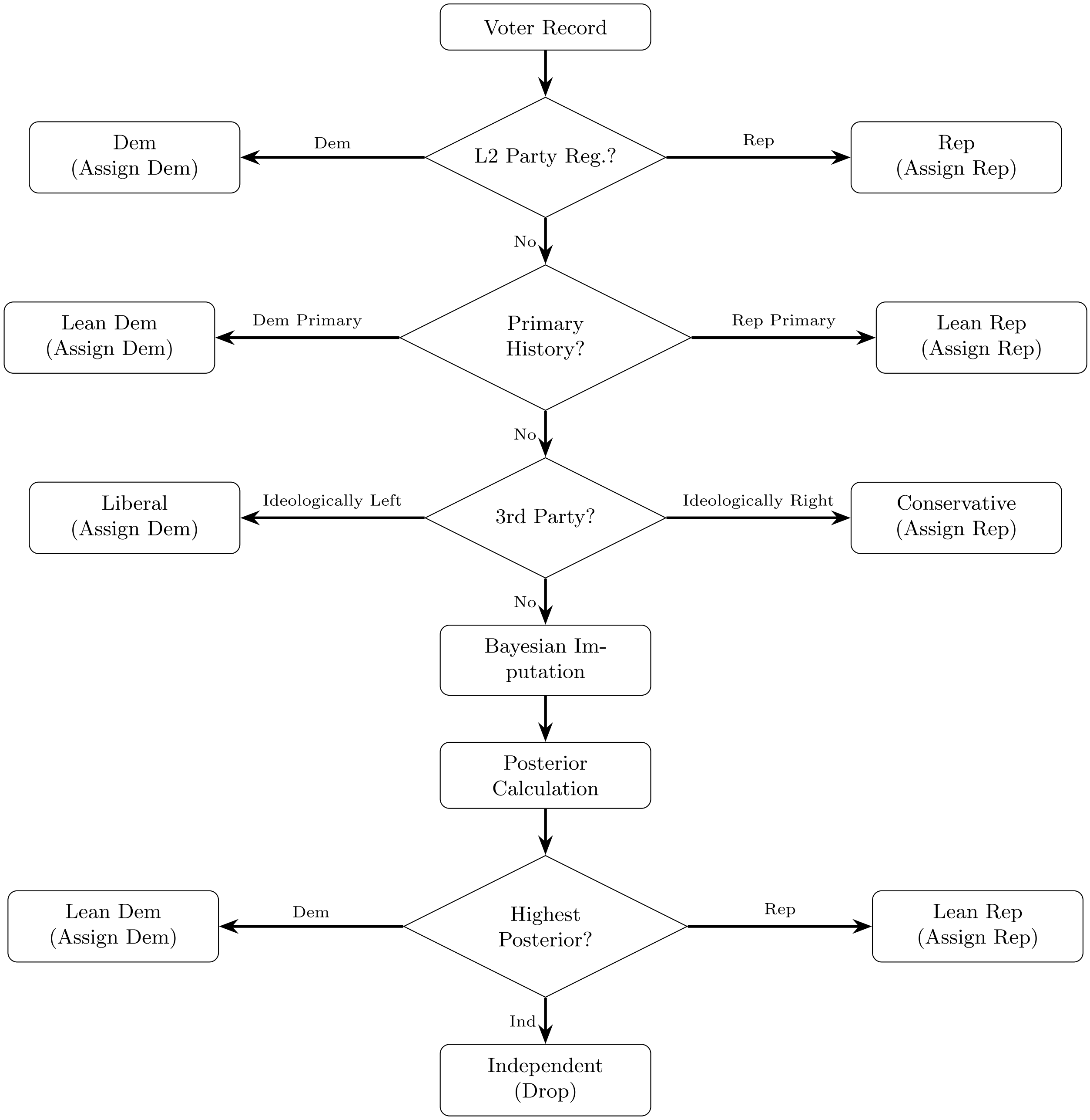

Extended Data Fig. 1 | Hierarchical partisanship imputation flowchart. Flowchart of the sequential rules used to assign each voter record a partisan affiliation (Democrat, Republican, or Independent). The algorithm first uses L2 party registration where available; otherwise it relies on primary voting history; otherwise it maps third-party registrations to the ideologically aligned major party; for the remaining records it performs Bayesian imputation and assigns the partisan label with the highest posterior probability. Records classified as Independent are dropped from the analytical sample. Full methodological details are provided in Methods and Supplementary Information.

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Extended Data Table 1 | Sample demographics vs. 2023 Cooperative Election Study (CES)

	(1) Our Merged Sample	(2) Working Reg. Voters	(3) Working Gen. Pop.	(4) Gen. Pop.
Party				
Democrat	0.41	0.43	0.34	0.32
Other	0.30	0.25	0.39	0.40
Republican	0.29	0.32	0.27	0.28
Race/Ethnicity				
White	0.61	0.72	0.69	0.69
Nonwhite	0.39	0.28	0.31	0.31
Gender				
Woman	0.53	0.48	0.52	0.51
Man	0.47	0.51	0.47	0.48

NOTES: Columns (2)–(4) calculated using the 2023 Cooperative Election Study (CES), weighted to registered voters (cols 2–3) or general population (col 4). Respondents reporting full- or part-time work are included in cols 2 and 3. Our merged sample uses the full matched sample (Democrats, Republicans, Others/Independents) with unimputed L2 party registration; this differs from the analytical sample used elsewhere, which is restricted to imputed Democrats and Republicans. Race/ethnicity uses L2 categories in our sample and self-report elsewhere; records with missing modeled race/ethnicity are excluded from race calculations only. Gender does not sum to 100% because of non-binary responses in the CES and a small share of missing gender in our sample.

Benchmarks our merged sample against the 2023 CES weighted to either registered voters (columns 2-3) or the general population (column 4). The merged sample uses unimputed L2 party registration over the full matched sample (Democrats, Republicans, Others/Independents). Race/ethnicity uses L2 categories in our sample and self-report elsewhere; records with missing modeled race/ethnicity are excluded from race calculations only. Gender does not sum to 100% because of non-binary responses in the CES and a small share of missing gender in our sample.

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Extended Data Table 2 | Summary statistics: Political workplace segregation analysis

Variable	N	Mean	Std. Dev.	Min	Max
Sample Characteristics					
Total Positions	37,200,614				
Unique Individuals	30,965,666				
MSAs	366				
Counties	1,392				
Census Tracts	72,252				
Workplace Characteristics					
Occupations	383				
Industries	1,013				
Workplace Size (MSA-level)	37,200,614	904	2,504	2	27,257
Workplace and Geographic Shares					
Republican Coworker Share (Two-Party)	37,200,614	0.406	0.239	0.000	1.000
Democratic Coworker Share (Two-Party)	37,200,614	0.586	0.242	0.000	1.000
Female Coworker Share (Benchmark)	37,200,614	0.530	0.253	0.000	1.000
Residential Republican Share	36,247,950	0.429	0.216	0.000	1.000
Individual Characteristics					
Republican (imputed)	15,235,660	0.410	0.492	0.000	1.000
Democrat (imputed)	21,964,961	0.590	0.492	0.000	1.000
Woman	19,896,974	0.535	0.499	0.000	1.000
Non-White	14,564,332	0.392	0.488	0.000	1.000
Age (2020)	36,845,511	42.8	16.5	14	96
Education					
High School or Lower	19,975,345	0.537	0.499	0.000	1.000
Bachelor's Degree	11,425,430	0.307	0.461	0.000	1.000
Graduate Degree	5,799,839	0.156	0.363	0.000	1.000
Seniority Levels					
Seniority Level 1	14,004,717	0.376	0.484	0.000	1.000
Seniority Level 2	9,320,556	0.251	0.433	0.000	1.000
Seniority Level 3	3,856,483	0.104	0.305	0.000	1.000
Seniority Level 4	3,861,352	0.104	0.305	0.000	1.000
Seniority Level 5	4,614,299	0.124	0.330	0.000	1.000
Seniority Level 6	1,169,624	0.031	0.175	0.000	1.000
Seniority Level 7	373,583	0.010	0.100	0.000	1.000
Political Activity					
2020 General Election	30,353,382	0.816	0.388	0.000	1.000
2020 Presidential Primary Voter	11,206,016	0.301	0.459	0.000	1.000
FEC Donor	1,316,417	0.035	0.185	0.000	1.000

NOTES: Summary statistics for the political workplace segregation analysis dataset used across Figures 1–5 and 7. Sample restricted to Democrats and Republicans (imputed partisanship) in establishments with 2 or more employees.

Reports counts (total positions, unique individuals, MSAs, counties, census tracts, occupations, industries), distributional statistics for workplace size, partisan and demographic coworker shares, individual characteristics (age, education, seniority), and political activity measures (2020 general-election turnout, 2020 presidential primary participation, FEC donor status).

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Extended Data Table 3 | Industry coverage vs. 2023 American Community Survey (ACS)

Supersector (NAICS Sector #)	Merged Sample	2023 ACS	Difference
Education and Health Services (61, 62)	0.20	0.23	-0.03
Financial Activities (52, 53)	0.15	0.07	+0.08
Manufacturing (31-33)	0.14	0.10	+0.04
Professional and Business Services (54-56)	0.13	0.12	+0.01
Public Administration (92)	0.09	0.05	+0.04
Information (51)	0.08	0.02	+0.06
Retail Trade (44-45)	0.08	0.11	-0.03
Leisure and Hospitality (71, 72)	0.04	0.09	-0.05
Transportation, Warehousing, and Utilities (22, 48-49)	0.03	0.06	-0.03
Wholesale Trade (42)	0.03	0.02	+0.01
Other Services (except Public Administration) (81)	0.02	0.05	-0.03
Construction (23)	0.02	0.07	-0.05
Natural Resources and Mining (11 and 21)	0.01	0.02	-0.01

NOTES: Population totals from 2023 American Community Survey (ACS) DP03 tables. Differences may reflect sample coverage but also differences in underlying populations: our merged sample aims to cover registered voters; the ACS covers the entire population including non-registered and non-citizens.

Share of the merged sample in each NAICS supersector compared to 2023 ACS DP03 tables, with differences in percentage points. Professional sectors where LinkedIn use is common (Financial Activities, Information, Manufacturing) are over-represented; Construction, Leisure and Hospitality, and Retail Trade are under-represented. Differences reflect both sample coverage and differences in the underlying population measured (registered voters vs. the ACS full resident population including non-citizens).

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Extended Data Table 4 | Coverage by occupation type (O*NET zones) vs. 2023 Bureau of Labor Statistics (BLS)

O*NET Zone	Merged Sample	Population	Difference
Zone One	0.02	0.07	-0.05
Zone Two	0.21	0.42	-0.21
Zone Three	0.22	0.22	+0.00
Zone Four	0.41	0.25	+0.16
Zone Five	0.14	0.05	+0.09

NOTES: Population figures from Bureau of Labor Statistics (BLS) May 2023 data release, merged with O*NET Zone codes from onetonline.org. O*NET Zones group occupations by required education, experience, and on-the-job training, and are used in government reporting and official purposes (e.g., US immigration visa eligibility).

Coverage across the five O*NET Job Zones benchmarked to May 2023 BLS Occupational Employment and Wage Statistics. O*NET Zones group occupations by required education, experience, and on-the-job training, and are used in government reporting (for example, US immigration visa eligibility). The merged sample over-represents higher-preparation zones (Zones 4 and 5) and under-represents lower-preparation zones (Zones 1 and 2).

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☒	☐	For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
☐	☒	Estimates of effect sizes (e.g. Cohen's d, Pearson's r), indicating how they were calculated

Software and code

Data collection	R 4.4.3; Splink v3.9.8 (probabilistic record linkage); FuzzyLink (R, LLM-based semantic matching); nominally (Python, name standardization). L2 voter file extracted November 2024. Revelio Labs employment data for positions active January 2024 through April 2025.
Data analysis	R 4.4.3. Key packages: fixest 0.13.2, arrow 21.0.0.1, ggplot2 3.5.2, dplyr 1.1.4, data.table 1.17.8, broom 1.0.8, readr 2.1.5, purrr 1.0.4, tidyr 1.3.1, ggridges 0.5.6, viridis 0.6.5.

Data

Proprietary data from Revelio Labs (revelio.com) and L2 Data (l2inc.com), both constructed from publicly available information. Due to licensing restrictions, individual-level data cannot be shared publicly. Synthetic replication data deposited at Harvard Dataverse: DOI 10.7910/DVN/1K7OUZ. Interested researchers can contact Revelio Labs and L2 Data directly for data access.

Research involving human participants, their data, or biological material

Reporting on sex and gender	Gender classification combines direct collection from 27 US states' voter registration files and probabilistic imputation for remaining states. Binary classification (men/women) used throughout. Gender variable sourced from L2 voter file.

Reporting on race, ethnicity, or other socially relevant groupings	Race/ethnicity from L2 voter files. Six Southern states collect race directly; all others use Bayesian Improved Surname Geocoding (BISG). Categories: non-Hispanic White, Hispanic, Asian, Black, Other. Race/ethnicity used as a covariate in heterogeneity analyses.
Population characteristics	37.2 million positions held by 31.0 million unique workers across 366 US MSAs. 53.5% women, 39.2% non-White, mean age 42.8 years. Partisan composition: 59.0% Democrat, 41.0% Republican (imputed).
Recruitment	Not applicable. This study uses administrative data (voter registration records and employment profiles). No human participants were recruited or contacted.
Ethics oversight	Reviewed by the Health Sciences and Behavioral Sciences IRB (IRB-HSBS) at the University of Michigan and determined to be 'Not Regulated' (HUM00258211, determined 7/22/2024).

Field-specific reporting

☐ Life sciences ☒ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

Behavioural & social sciences study design

All studies must disclose on these points even when the disclosure is negative.

Study description	Observational, cross-sectional and longitudinal study examining political workplace segregation using linked administrative data on employment and voter registration across 366 US metropolitan areas.
Research sample	37,200,614 employment positions held by 30,965,666 unique workers across 366 US MSAs. 53.5% women, 39.2% non-White, mean age 42.8 years. 59.0% Democrat, 41.0% Republican (imputed partisanship). Sample is not fully representative: over-represents white-collar industries and higher-education occupations due to reliance on LinkedIn-derived employment data. Benchmarked against the 2023 Cooperative Election Study.
Sampling strategy	Universe of matched records between Revelio Labs employment data (sourced from LinkedIn profiles and online job postings) and L2 voter registration files. Ensemble matching using probabilistic record linkage (Splink) and LLM-based semantic matching (FuzzyLink).
Data collection	L2 voter file: extracted and processed November 2024. Revelio Labs employment data: positions active January 2024 through April 2025. Panel dataset extends 2012-2024 using historical position data. No primary data collection; all data are from existing administrative sources.
Timing	L2 voter file: November 2024. Revelio Labs: positions active January 2024 to April 2025. Longitudinal panel: 2012-2024. Primary election history used for imputation: 2012-2022. Geographic priors based on 2020 presidential election precinct returns.
Data exclusions	Documented in Methods (Sample Attrition) and Supplementary Figures 11-12. From the matched sample, the following restrictions are applied: (1) non-missing values for partisan exposure measures, imputed partisanship, gender, and age; (2) workplaces with at least 2 matched employees; (3) valid industry (NAICS) and occupation (O*NET-SOC) classifications; (4) restriction to workers classified as Democrats or Republicans via imputed partisanship, excluding independents and third-party registrants; (5) valid geocoded residential addresses for census tract assignment. All exclusion criteria were established before analysis.
Non-participation	Not applicable. This study uses administrative data; there was no recruitment of participants. Match rates between datasets vary by age and are documented in Supplementary Figure 1.
Randomization	Not applicable. This is an observational study; no experimental randomization was performed.